AMERICAN SOCIETY FOR MICROBIOLOGY | Applied and Environmental Microbiology®

ENVIRONMENTAL MICROBIOLOGY

Transcriptional Control of *hgcAB* by an ArsR-Like Regulator in *Pseudodesulfovibrio mercurii* ND132

Caitlin M. Gionfriddo,[a]* Ally Bullock Soren,[a] Ann M. Wymore,[b] D. Sean Hartnett,[a] Mircea Podar,[b] Jerry M. Parks,[b] Dwayne A. Elias,[c] Cynthia C. Gilmour[a]

[a]Smithsonian Environmental Research Center, Edgewater, Maryland, USA
[b]Biosciences Division, Oak Ridge National Lab, Oak Ridge, Tennessee, USA
[c]Elias Consulting, LLC, Knoxville, Tennessee, USA

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

ABSTRACT The *hgcAB* gene pair encodes mercury (Hg) methylation capability in a diverse group of microorganisms, but its evolution and transcriptional regulation remain unknown. Working from the possibility that the evolutionary function of HgcAB may not be Hg methylation, we test a possible link to arsenic resistance. Using model Hg methylator *Pseudodesulfovibrio mercurii* ND132, we evaluated transcriptional control of *hgcAB* by a putative ArsR encoded upstream and cotranscribed with *hgcAB*. This regulator shares homology with ArsR repressors of arsenic resistance and *S*-adenosylhomocysteine (SAH)-responsive regulators of methionine biosynthesis but is distinct from other ArsR/SahR proteins in *P. mercurii*. Using quantitative PCR (qPCR) and RNA sequencing (RNA-seq) transcriptome analyses, we confirmed this ArsR regulates *hgcAB* transcription and is responsive to arsenic and SAH. Additionally, RNA-seq indicated a possible link between *hgcAB* activity and arsenic transformations, with significant upregulation of other ArsR-regulated arsenic resistance operons alongside *hgcAB*. Interestingly, wild-type ND132 was less sensitive to As(V) (but not As(III)) than an *hgcAB* knockout strain, supporting the idea that *hgcAB* may be linked to arsenic resistance. Arsenic significantly impacted rates of Hg methylation by ND132; however, responses varied with culture conditions. Differences in growth and metabolic activity did not account for arsenic impacts on methylation. While arsenic significantly increased *hgcAB* expression, *hgcAB* gene and transcript abundance was not a good predictor of Hg methylation rates. Taken together, these results support the idea that Hg and As cycling are linked in *P. mercurii* ND132. Our results may hold clues to the evolution of *hgcAB* and the controls on Hg methylation in nature.

IMPORTANCE This work reveals a link between microbial mercury methylation and arsenic resistance and may hold clues to the evolution of mercury methylation genes (*hgcAB*). Microbes with *hgcAB* produce methylmercury, a strong neurotoxin that readily accumulates in the food web. This study addresses a critical gap in our understanding about the environmental factors that control *hgcAB* expression. We show that *hgcAB* expression is controlled by an ArsR-like regulator responsive to both arsenic and *S*-adenosylhomocysteine in our model organism, *Pseudodesulfovibrio mercurii* ND132. Exposure to arsenic also significantly impacted *Pseudodesulfovibrio mercurii* ND132 mercury methylation rates. However, expression of *hgcAB* was not always a good predictor of Hg methylation rates, highlighting the roles of Hg bioavailability and other biochemical mechanisms in methylmercury production. This study improves our understanding of the controls on *hgcAB* expression, which is needed to better predict environmental methylmercury production.

KEYWORDS mercury biogeochemistry, arsenic resistance, ArsR, *hgcAB*, methylmercury, RNA-seq, *ars* operon, arsenic, mercury

Editor Nicole R. Buan, University of Nebraska-Lincoln

This is a work of the U.S. Government and is not subject to copyright protection in the United States. Foreign copyrights may apply.

Address correspondence to Caitlin M. Gionfriddo, c.gionfriddo@latrobe.edu.au.

*Present address: Caitlin M. Gionfriddo, Centre for Freshwater Ecosystems, School of Agriculture, Biomedicine and Environment, La Trobe University, Wodonga, Victoria, Australia.

The authors declare no conflict of interest.

Received 14 October 2022
Accepted 21 February 2023
Published 23 March 2023

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

Microorganisms play a major role in controlling the form and fate of mercury (Hg) in aquatic environments. Methylmercury (MeHg), an organic Hg compound, is a neurotoxin that accumulates in food sources and is primarily produced by microorganisms (1). The *hgcAB* gene pair that encodes Hg methylation capability (2) is widespread in anaerobic environments and present in genomes spanning a diverse group of metabolic clades (3–5). Hg methylation capability is a species-specific trait that is often associated with low-abundance taxa in the microbiome (3, 6). Currently, we do not have a good metric for estimating the prevalence of the Hg methylation trait among microbiota. However, in environmental metagenomic studies of Hg methylator environments, sequencing coverage of *hgcAB* genes is often low compared to that of other functional genes (7). For example, the coverage of dissimilatory sulfite reductase genes (*dsr*) in metagenomic assemblies is often 3 to 34 times higher than that of *hgcAB* (7). Despite the association of sulfate-reducing bacteria (SRB) with Hg methylation, the *hgcAB* gene pair is found in only ~12% of publicly sequenced SRB in the *Deltaproteobacteria* (although *hgcAB* is more common within some genera, including *Pseudodesulfovibrio*) (8–10). Of the roughly 3,078 sequenced *Deltaproteobacteria* genomes in the NCBI reference database, only 368 were identified as potential Hg methylators when the Hg-MATE database was compiled (7, 11). Within other phyla, especially environmental metagenome-assembled genomes (MAGs) from the *Bacteroidetes*, *Nitrospirae*, PVC superphylum, and candidate phyla radiation, *hgcAB* distribution further diverges from species trees but follows environmental patterns and suggests horizontal gene transfer of this capability (4, 6). Not all Hg methylators share the same methylation capabilities, indicating that Hg uptake, Hg(II) binding to HgcAB, *hgcAB* transcription, enzyme regulation, or protein structure may control Hg methylation capacity and efficiency (12–17). Currently, we have a limited understanding of the biochemical constraints on Hg methylation and the environmental conditions that resulted in the conservation of *hgcAB* in bacterial and archaeal genomes.

Despite the identification of two genes that encode Hg methylation capability, the mechanism by which methylation is performed within the cell is largely speculative. The genes encode a corrinoid (i.e., vitamin B_{12}-dependent) protein (HgcA), and a 2[4Fe-4S] ferredoxin (HgcB), which could be used as a methyl carrier and electron donor in corrinoid cofactor reduction, respectively (2, 18). Both genes (*hgcA* and *hgcB*) are required for Hg methylation (2, 19). Corrinoid-dependent methyltransferases are utilized in a range of microbial metabolic pathways in the methylation of amino acids, proteins, DNA, and other intermediates (20, 21). In anaerobic microorganisms, corrinoid-dependent enzymes drive methionine and acetyl coenzyme A (acetyl-CoA) production (22, 23). Both pathways have been linked to Hg methylation, and the methyl transfer to Hg is thought to occur via a similar mechanism (18, 24–27). HgcA consists of a cytoplasmic corrinoid binding domain and a transmembrane domain. The cytoplasmic domain of HgcA shares homology with the C-terminal domain of the large subunit of the corrinoid iron-sulfur protein (CFeSP) utilized in methyl transfer in the Wood-Ljungdahl (i.e., reductive acetyl-CoA) pathway (2, 18).

Further, the native function(s) of HgcA and HgcB remains unknown. Although HgcAB function has been linked to one-carbon metabolism (28), we do not know if these proteins carry out any functions independent of Hg methylation. Redox transformations of metals and metalloids are often part of detoxification mechanisms or energy production. However, biomethylation of Hg does not provide any observed benefit to the cell, and Hg methylation capability is not induced by Hg(II) (28–30). Deletion of the *hgcAB* gene pair does not appear to hinder cell growth or produce any major phenotype in *Pseudodesulfovibrio mercurii* ND132 (2, 28). While some photosynthetic bacteria have been shown to use Hg(II) as an electron acceptor during photosynthetic and fermentative growth (31), Hg-methylating organisms are not able to use Hg species as growth substrates (30). Resolving the native biochemical function of HgcAB is needed to identify the metabolic conditions that mediate HgcAB activity and function. Understanding the biological and geochemical factors that influence MeHg

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

production is necessary for predicting the impacts of contamination and remediation efforts on natural systems.

Building predictive models for net MeHg production in aquatic environments is a challenge (32). There are several microbial, physical, and geochemical parameters that influence Hg biogeochemistry in aquatic environments, including but not limited to Hg complexation, availability of terminal electron acceptors, shifts in redox, the composition or availability of organic matter, and the presence, abundance, and activity of Hg-cycling microorganisms (1, 33, 34). While the *hgcAB* gene pair can be used to identify potential Hg methylators in a microbiome (6), to date, *hgcAB* gene abundance alone has rarely provided a useful indicator of environmental MeHg production (35–38). Measuring metabolic and *hgcAB* activity of Hg methylators has been proposed to improve predictions (36, 39–42). However, it is unclear whether we can relate *hgcA* activity to Hg methylation rates (29). There are several significant unknowns, including whether *hgcAB* is transcriptionally regulated, what geochemical conditions influence *hgcAB* transcription, and whether higher expression of *hgcAB* transcripts increases MeHg production. Previous work indicates that *hgcAB* transcription is not regulated by Hg (28–30). A possible overlap with microbial mechanisms for arsenic resistance has been proposed, as several studies have shown that *hgcAB* is colocalized with genes that encode arsenic resistance in some Hg methylators (4, 8, 43), including an ArsR-like regulator. In Hg-methylating members of the *Pseudodesulfovibrio* genus, this *arsR*-like gene is often found with *hgcAB* and has been shown to be cotranscribed with *hgcAB* (8).

In this study, we test the ability of the ArsR-like regulator cotranscribed with *hgcAB* to enhance both *hgcA* transcription and Hg methylation itself in *Pseudodesulfovibrio mercurii* ND132, a model Hg methylator (9, 30). Two transcriptional responses were tested, one to arsenic using arsenite [As(III)] and arsenate [As(V)] and the second to *S*-adenosylhomocysteine (SAH) using methionine and adenosine-2′,3′-dialdehyde. We chose these test conditions due to the shared homology between this potential *hgcAB* regulator and helix-turn-helix (HTH) regulons in other sulfate-reducing bacteria that are responsive to As(III) (ArsR) and *S*-adenosylhomocysteine (SahR). These regulate arsenic resistance or methionine biosynthesis, respectively (44–46). There are possible biochemical overlaps between Hg methylation and *S*-adenosylmethionine (SAM)-mediated methylation (47, 48), including those utilized in methyl-arsenic or methionine biosynthesis. Experiments were designed to test whether *hgcAB* transcription is regulated by ArsR/SahR.

In addition to evaluating potential *hgcA* regulators, another goal of this study was to test the idea that Hg methylation rates are related to *hgcAB* expression rates. We hypothesized that geochemical culture conditions that increase *hgcAB* expression in *Pseudodesulfovibrio mercurii* ND132 will lead to higher Hg methylation rates than analogous conditions. Outcomes of this study have important implications for modeling Hg methylation potential in the environment.

RESULTS AND DISCUSSION

An ArsR-like regulator encoded with HgcAB in *Pseudodesulfovibrio*. A putative arsenic-responsive regulator (ArsR) (gene identifier [ID] DND132_1054, UniProt accession no. F0JBE8) is encoded directly upstream of Hg methylation genes *hgcA* (gene ID DND132_1056, UniProt accession no. F0JBF0) and *hgcB* (gene ID DND132_1057, UniProt accession no. F0JBF1) in the genome of *P. mercurii* ND132. Potential ArsR-controlled transcriptional regulation of *hgcAB* has been proposed for Hg-methylating members of the *Pseudodesulfovibrio* genus (8). This potential *hgcAB* regulator shares homology to the metalloregulator ArsR/SmtB protein family (Pfam PF01022), including a DNA binding helix-turn-helix (HTH) motif. This family of regulators (Pfam PF01022) is responsive to metal ion stress [e.g., As(III), Zn(II), and Cd(II)] and in the absence of coordinating metal(loid)s will repress the expression of operons linked to heavy metal resistance (49). This transcriptional family includes the ArsR repressor of the *ars* operon, which often encodes an arsenate reductase with arsenic permeases and exporters (50). However, protein sequence analysis and predictive structure modeling of the *hgcAB* ArsR-like regulator

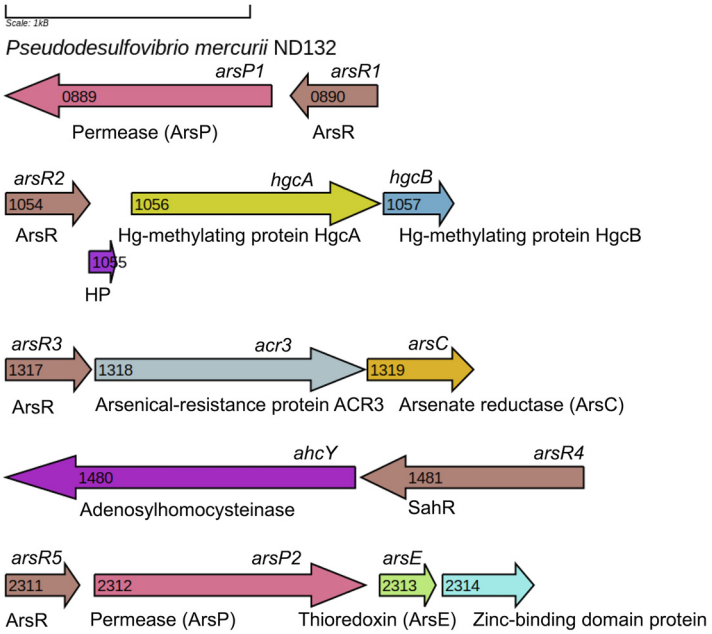


FIG 1 ArsR/SmtB family transcriptional regulators encoded in the genome of *Pseudodesulfovibrio mercurii* ND132. The five *arsR* genes are numbered by order of appearance in the genome, with gene IDs in arrows (e.g., HgcA encoded by DND132_1056) and gene orientation shown by arrows, and are color coded by protein family. The figure was made using Gene Graphics (96).

suggest differences in the metal binding domain that may make it distinct from other ArsR family regulators.

There are five ArsR family transcriptional regulators annotated in the genome of *Pseudodesulfovibrio mercurii* ND132 (Fig. 1). Based on protein sequence analysis, these five ArsR proteins are different from one another, with less than 33% pairwise identity between them (see Table S3 in the supplemental material). We numbered the *arsR* genes by order of appearance in the genome of *P. mercurii* ND132, and *arsR2* is located directly upstream of *hgcA* and *hgcB* (Fig. 1). While in some Hg-methylating bacteria *hgcAB* is located in the genome with arsenic resistance (*ars*) genes (4, 43), no other *ars* genes are colocalized with *hgcAB* in ND132. However, three of the other *arsR* genes (*arsR1*, *arsR3*, and *arsR5*) are present with *ars* genes elsewhere in the genome and likely regulate arsenic resistance. The first set of *ars* genes encodes ArsR1 (gene ID DND132_0890) and the first of two methylarsenite efflux permeases, ArsP (DND132_0889) (51). Another *ars* operon encodes ArsR3 (DND132_1317) with the first of two Acr3 family As(III) permeases (DND132_1318) and arsenate reductase, ArsC (DND 132_1319). *P. mercurii* ND132 has a second Acr3 family As(III) efflux permease that is not part of an *ars* operon (DND132_2196). The third *ars* operon is predicted to encode ArsR5 (DND132_2311), another ArsP (DND132_2312), an organoarsenical resistance protein, ArsE (Pfam PF13192, thioredoxin-like protein) (DND132_2313) (52), and a protein of unknown function with a diguanylate cyclase (DGC) motif (Pfam PF08859, a domain with four conserved cysteines that likely coordinate Zn) (DND132_2314). ArsR repressors of the *ars* operon are predicted to be transcriptionally regulated by As(III) and part of a detoxification pathway for arsenic. From protein sequence analysis of ND132 ArsR2, it is unclear whether expression of *hgcA* and *hgcB* is part of the ArsR regulon and controlled by arsenic.

ArsR/SmtB family regulons may not always be associated with heavy metal resistance or arsenic transformations. In addition to regulating arsenic resistance, ArsR regulators have also been shown to influence many other cellular functions in the presence of As(III), including phosphate acquisition/metabolism, sugar transport, chemotaxis, copper tolerance, and iron homeostasis (53). Regulators in the ArsR/SmtB HTH family may also be responsive to substrates other than heavy metals and metalloids. In

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

sulfate-reducing bacteria, an ArsR-like regulon responsive to intracellular concentrations of *S*-adenosylhomocysteine (SAH) controls SAM-dependent methyl transfer for methionine biosynthesis (44, 45). These SAH-responsive regulators (SahR) are often encoded with adenosylhomocysteinase (AHCY), which catalyzes the hydrolysis of SAH to homocysteine and adenosine. In *P. mercurii* ND132, *arsR4* likely encodes a SahR (gene ID DND132_1481, UniProt accession no. F0JE80). The predicted SahR is encoded with AHCY (gene ID DND132_1480, UniProt accession no. F0JE79) (Fig. 1) and has an ArsR-like HTH motif with a SAM-dependent methyltransferase similar to ArsM. Arsenite methyltransferase (ArsM) catalyzes the transfer of a methyl group to arsenite from *S*-adenosylmethionine (SAM) to form methylarsenicals and is part of the ArsR regulon (54). Instead of methylarsenical production, ND132 ArsR4 is predicted to control methionine production and be responsive to SAH. No other homologues of ArsM are encoded in the genome of *P. mercurii* ND132. Like *arsR2*, which is colocalized with *hgcAB*, *arsR4* is not part of an *ars* operon in *P. mercurii* ND132.

Phylogenetic distribution of the ArsR2 *hgcAB* regulator. Close homologues of the ArsR2 regulator from *Pseudodesulfovibrio mercurii* ND132 are present in other *hgc*$^+$ *Desulfovibrio* and *Pseudodesulfovibrio* genomes (Fig. S1). To identify potential ArsR2 homologues, we searched for ArsR family proteins encoded before *hgcAB* in *Desulfovibrio* and *Pseudodesulfovibrio* reference genomes from a previous study (9). We also used the hidden Markov model (HMM) of the HTH motif of ArsR (Pfam PF01022, HTH_5.hmm) to broaden our search and identify all potential HTH family transcriptional regulators encoded in the reference genomes. We used protein sequence alignments and maximum likelihood phylogenies of HMM hits to identify ArsR2 homologues. Among these genomes, we found ArsR2 homologues only in *hgc*$^+$ *Desulfovibrionaceae* genomes and always directly upstream of *hgcAB* (Fig. S1). Other researchers working with a close relative to ND132, *Pseudodesulfovibrio hydrargyri* BerOc1, demonstrated that these *arsR*-like genes and *hgcA* are coexpressed (8). Notably, all but one of the available *Pseudodesulfovibrio* genomes encode the ArsR2 homologue. However, cooccurrence of *arsR* with *hgcAB* does not appear to be phylogenetically conserved in the majority of *hgc*$^+$ genomes of close relatives of *Desulfovibrionaceae* (Fig. S1).

ArsR family regulators are also encoded with *hgcAB* in genomes from outside the *Desulfovibrionaceae* family. In *hgc*$^+$ genomes from the Hg-MATE-Db (7, 11), we identified putative ArsR/SmtB family regulators of *hgcAB* in isolate and environmental metagenome-assembled genomes (MAGs) from *Deltaproteobacteria*, *Nitrospira*, *Spirochaetes*, *Ignavibacteriae* (55), and *Bacteroidetes* (56), including *Paludibacter jiangxiensis* NM7 (57) (Table S4). Based on protein sequence alignments, not all of these putative *hgcAB* regulators are close homologues of the ArsR2 protein from *Pseudodesulfovibrio mercurii* ND132. In some of these genomes, *hgcA* and *hgcB* are also colocalized with other *ars* genes (e.g., *arsC*, *arsB*, and *arsM*). It is noteworthy that many of the *hgc*$^+$ MAGs (58) of sediment- and groundwater-associated bacteria from the Rifle IFRC site (59) have an ArsR-like regulator encoded with *hgcAB* and *ars* genes. This cooccurrence of *hgcAB* with *ars* genes in MAGs from arsenic-impacted environments has been noted previously (4, 43). For this study, we focused our analytical efforts on understanding possible ArsR-regulated transcription of *hgcAB* in *Pseudodesulfovibrio mercurii* ND132. However, more analysis is needed to explore a possible evolutionary relationship between the *ars* operon and Hg methylation genes and whether horizontal gene transfer of *hgcAB* may have been influenced by arsenic.

Structural model. The ArsR2 homologues from *Pseudodesulfovibrio* and *Desulfovibrio* are phylogenetically distinct from other ArsR/SmtB family transcriptional regulators (Fig. S2). To date, the function of these ArsR2 homologues has not been experimentally confirmed through gene deletion, and they are therefore referred to as putative As or *hgcAB* regulators. The protein sequence alignment shows that ArsR2 has some of the conserved invariant residues associated with the HTH family of regulators (Fig. S2) (60). To provide additional insight into the structure of the putative *hgcAB* transcriptional regulator (ArsR2, DND132_1054), we generated a homodimeric model of the protein using AlphaFold2 (61, 62) (Fig. 2). The cysteine-rich site associated with metal(loid) binding is different in ArsR2 than in ArsR repressors of the *ars* operon. In *Escherichia coli* ArsR

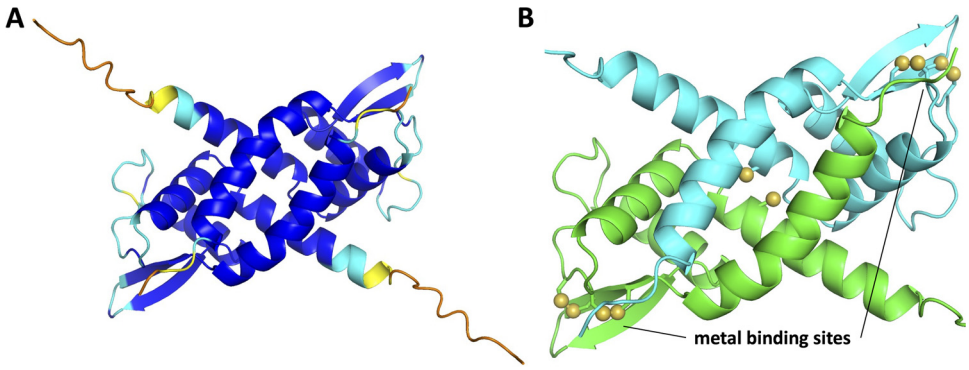


FIG 2 (A) AlphaFold2 model of the putative ArsR-like transcriptional regulator encoded by *arsR2* (UniProt ID F0JBE8) and colocalized with *hgcAB* in the genome of *Pseudodesulfovibrio mercurii* ND132. The model is colored by model confidence (i.e., pLDDT score). Dark blue, very high confidence (pLDDT > 90); cyan, confidence (90 > pLDDT > 70); yellow, low confidence (70 > pLDDT > 50); orange, very low confidence (pLDDT < 50). (B) Close-up view colored by chain with cysteine residues shown in the ball-and-stick representation.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

(UniProt accession no. P15905), the As(III) binding site is associated with the three cysteines on the α3 helix (Cys32, Cys34, and Cys37) (49). The cysteines in ND132 ArsR2 (UniProt accession no. F0JBE8) occur on a coil after helix 2 and a beta strand before helix 4 (Fig. S3). The ArsR2 from ND132 includes a cysteine (Cys26) near the N terminus and four cysteines, Cys47, Cys49, Cys86, and Cys88 (Fig. 2 and Fig. S3), in each predicted metal(loid) binding site (60). Homologues from several other bacteria closely related to *Pseudodesulfovibrio* and *Desulfovibrio* have similar structures and cysteine positions (Fig. S4), suggesting that some of them may have similar functions. The four-cysteine motif in the metal(loid) binding site is conserved in all ArsR2 homologues found in *Pseudodesulfovibrio* and *Desulfovibrio* genomes (Fig. S4), except for *Desulfovibrio brasiliensis* (WP_157054761.1), which has a three-cysteine motif.

The metal(loid) binding domain of the putative ArsR of *hgcAB* in *Pseudodesulfovibrio mercurii* ND132 (ArsR2, DND132_1054) differs from that in other ArsR repressors, containing four rather than three cysteines. Arsenite [$As(OH)_3$] is predicted to coordinate with the three cysteine residues of ArsR as As(III), after the loss of its hydroxide ligands (60). However, only two of the three cysteines (Cys32 and Cys34) are needed to bind arsenic to ArsR of *Escherichia coli* plasmid R773 (63). The presence of four cysteine residues in the metal(loid) binding sites of the ArsR2 regulator suggests it may preferentially bind metal(loid) cations such as Zn(II), Cd(II), and Pb(II) like the four cysteine residues in the binding motif of CadC (60). Alternatively, there could be 3-coordinate binding of As(III) to three of the four cysteine residues as has been proposed with As(III) and the SAM methyltransferase, ArsM (64).

Arsenic tolerance in ND132. The presence of a putative ArsR regulator upstream of *hgcAB* suggests that the ability to methylate Hg may confer As resistance in some microbes. We compared As toxicity between wild-type (WT) ND132 and the *hgcAB* deletion mutant (ND132Δ*hgcAB*) to test the idea that the presence of *hgcAB* in the ND132 genome might contribute to As resistance. We performed growth assays in defined (no yeast extract) CCM-PF (pyruvate-fumarate) and CCM-LS (lactate-sulfate) media. Cultures were inoculated into medium amended with sodium arsenate (0.2, 0.5, 1, 2, 5, or 20 mM) or sodium arsenite (0.02, 0.2, or 0.5 mM) and grown at 30°C to stationary phase (Fig. 3 and Fig. S5).

The WT strain was more tolerant of As(V) than the knockout. Specifically, WT ND132 growth was unaffected by As(V) levels up to 1 mM on both media, while growth of ND132Δ*hgcAB* was significantly impacted at that level (Fig. 3). Results were evaluated statistically by normalizing measurements of optical density at 600 nm (OD_{600}) to matched unamended controls and then evaluating differences in normalized OD_{600}s between the strains. Data from the exponential part of the growth curves were used in the statistical tests. Differences were assessed using general linear models that accounted

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

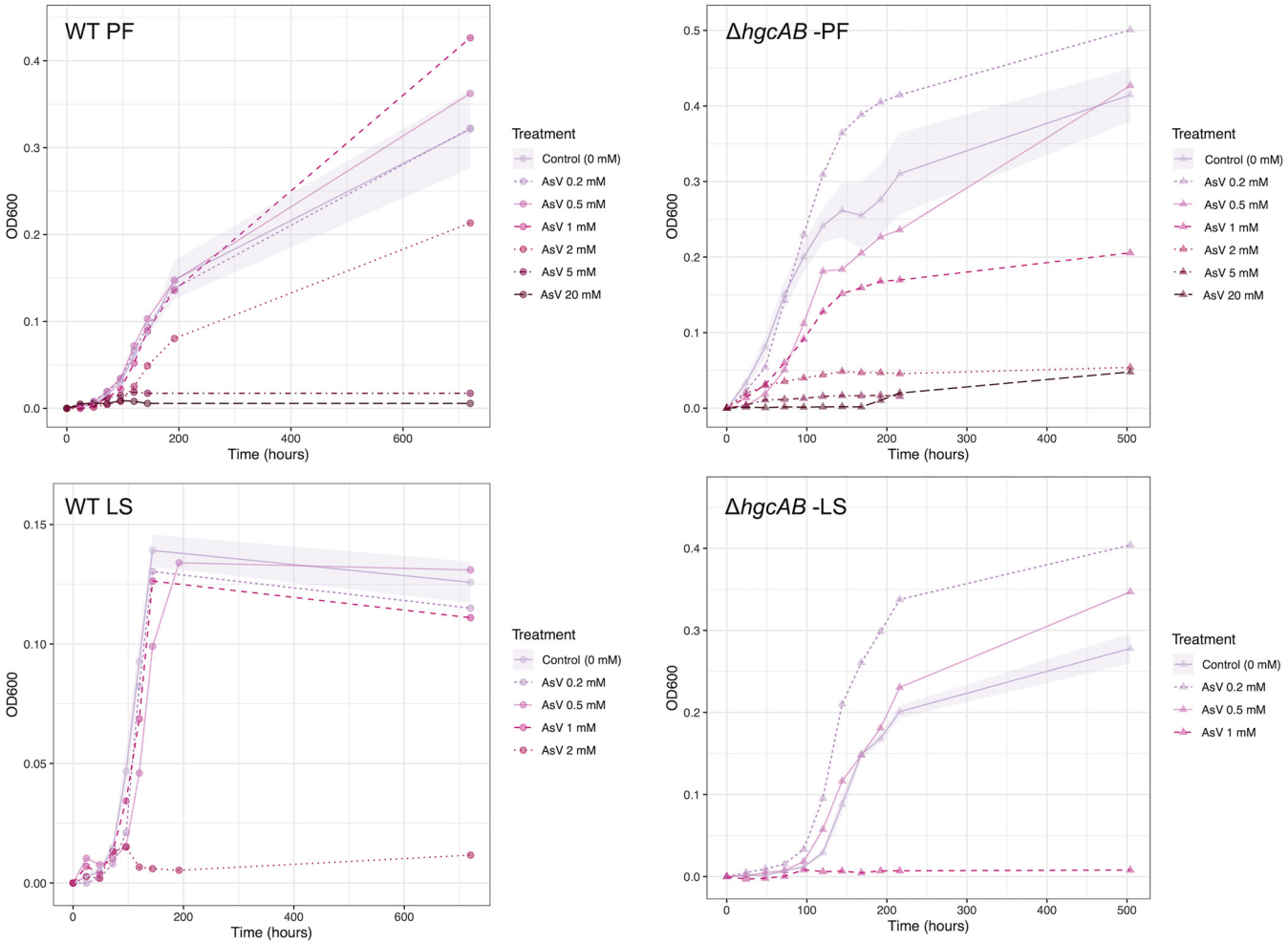


FIG 3 Growth assay tests of arsenate toxicity to *Pseudodesulfovibrio mercurii* ND132 wild type (WT) and *hgcAB* deletion mutant (Δ*hgcAB*) in defined CCM medium with pyruvate-fumarate (PF) or lactate-sulfate (LS) and sodium arsenate, As(V). Growth curves fitted to optical density measured at 600-nm absorbance (OD_{600}) of biological triplicates using 'ggplot' and 'dplyr' in R. Shaded area shows the standard error for average OD_{600} measurements for control replicates.

for As concentration, time, and any significant interactions. The *hgcAB* knockout was significantly more sensitive to As(V) at levels above 0.2 mM on PF medium and at levels above 0.5 mM on LS medium ($P < 0.0001$). Notably, there was apparent hormesis in the growth of ND132Δ*hgcAB*, with slightly enhanced growth at the lowest tested As(V) levels.

The WT and knockout strains were equally resistant to As(III), although growth was completely inhibited at As(III) concentrations roughly 4 to 10 times lower than those of As (V) (Fig. S5). The higher tolerance of *P. mercurii* ND132 to As(V) than to As(III) is consistent with other microorganisms, as As(III) is often more toxic than As(V) in most biological systems (65). Culture chemistry, particularly the formation of As-sulfide complexes, impacted As toxicity. Both wild-type and ND132Δ*hgcAB* cultures had lower tolerance to both As species when grown under sulfidogenic conditions, with lactate and sulfate, than when grown with pyruvate and fumarate as the growth substrates. A yellow mineral precipitate was observed when medium with 0.5 mM As(III) was inoculated with sulfidogenic *P. mercurii* ND132 WT and Δ*hgcAB* cultures and likely was the formation of orpiment (As_2S_3).

The reduced As(V) resistance of the *hgcAB* knockout strain suggests that *hgcAB* may confer some As(V) resistance. Alternatively, the loss of *hgcAB* from the genome may impact other mechanisms of As(V) resistance by ND132. This observation deserves more exploration, especially the lack of differential resistance to the more toxic As(III).

ND132 transformation of As(III) and As(V). *Pseudodesulfovibrio mercurii* ND132 has several ArsR-regulated arsenic resistance operons (Fig. 1). These operons are predicted

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

to encode As(V) reductase, As(III) and methylarsenite permeases, and a putative organic arsenic resistance protein. To determine how ND132 may transform arsenic (As), we measured As speciation and mass balance in *P. mercurii* ND132 WT cultures grown with either 0.02 mM sodium arsenite or 0.2 mM sodium arsenate in defined CCM-PF medium. At these concentrations, As did not affect cell growth rate (Fig. S6A). We did not examine whether ND132 gains energy from As(V) reduction.

Under these conditions, we found that wild-type *P. mercurii* ND132 reduced As(V) to As(III) and produced potential thioarsenicals from both As(III) and As(V) (Fig. S6 and Table S5). As concentrations and speciation were measured initially and at the end of exponential growth (7 days). ND132 reduced about 25% of added As(V) to As(III) after 7 days and converted about 3% to an unidentified As compound. This compound was tentatively identified as a thioarsenical based on the retention time of the peak (B. Wozniak, Brooks Applied Labs, personal communication). ND132 converted a higher fraction (roughly 15%) of added As(III) to the presumed thioarsenical compound (Fig. S6).

In both As(III)- and As(V)-amended cultures, As speciation at the initial time point matched the spikes, and 70 to 80% of the added As could be accounted for after 7 days of growth. Uninoculated medium served as abiotic controls. There was no significant loss of As or change in As speciation in abiotic controls, including no production of unidentified As compounds. No monomethyl- or dimethylarsenicals were detected in abiotic or cultured samples.

Potential transcriptional regulation of *hgcA* by ArsR2. To help determine the transcriptional controls on the putative ArsR regulator of *hgcAB*, we challenged cells with As and SAH and evaluated expression of both *arsR2* and *hgcA*. We chose these potential regulators because, despite shared homology, the putative *hgcAB* regulator ArsR2 is distinct from other HTH regulators of the ArsR/SmtB family, and its function is unknown. Additionally, since the four other ArsR proteins annotated in the genome of *P. mercurii* ND132 are predicted to regulate either methionine production or As resistance, we speculate that there could be biochemical overlaps between HgcAB function and As resistance or methionine production. We made this inference based on the colocalization of *hgcAB* with other *ars* genes (4, 43) and the shared homology of HgcA/B to corrinoid iron-sulfur proteins utilized in one-carbon metabolism (18, 66), including methionine biosynthesis. Thus, one goal of this work was to test the idea that *hgcAB* may be part of the ArsR or SahR regulon. Because of the unusual configuration of the metal(loid) binding motif in ArsR2, we chose to test the ND132 response to As(V) as well as As(III) and SAH.

We used real-time quantitative PCR (qPCR) to test relative changes in expression of *arsR2* and *hgcA* during growth with potential transcription regulators [e.g., As(III), As(V), and SAH]. Tests were conducted in defined (i.e., no yeast) CCM medium or yeast-enriched estuarine pyruvate fumarate (EPF) medium. Expression was normalized to three housekeeping genes and compared to control cultures without added treatments. We also tested for an effect of Hg on *hgcA* transcription and found none (Fig. 4). This observation is consistent with previous work that *hgcA* expression and Hg methylation activity are not induced by Hg (28–30). To evaluate a potential *arsR2* transcriptional response to SAH, we tested several compounds in the methionine biosynthesis pathway. Because *P. mercurii* ND132 is not predicted to have an SAH uptake mechanism, we tested for transcriptional response to SAH by growing cells with methionine (1 mM), adenosine-2′,3′-dialdehyde (0.1 mM), and equimolar homocysteine with adenosine (0.2/0.2 mM, 0.5/0.5 mM, and 1/1 mM), which enabled the buildup of SAH in the cell. Adenosine-2′,3′-dialdehyde is an inhibitor of SAM-dependent methyltransferase. Methionine, homocysteine, and adenosine are metabolites in the SAM cycle, and higher concentrations in culture medium should contribute to higher concentrations of SAH than in cultures grown without these substrates. Growth of *P. mercurii* ND132 was completely inhibited at all concentrations of homocysteine with adenosine. Therefore, this treatment was not used for further expression or Hg methylation experiments.

Transcription of *hgcA* is controlled by arsenic and SAH. We found that ArsR2 regulated *hgcA* expression in *P. mercurii* ND132 and can be responsive to both As and SAH. We observed significantly higher expression (z-test, one-tailed *P* value of <0.05

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

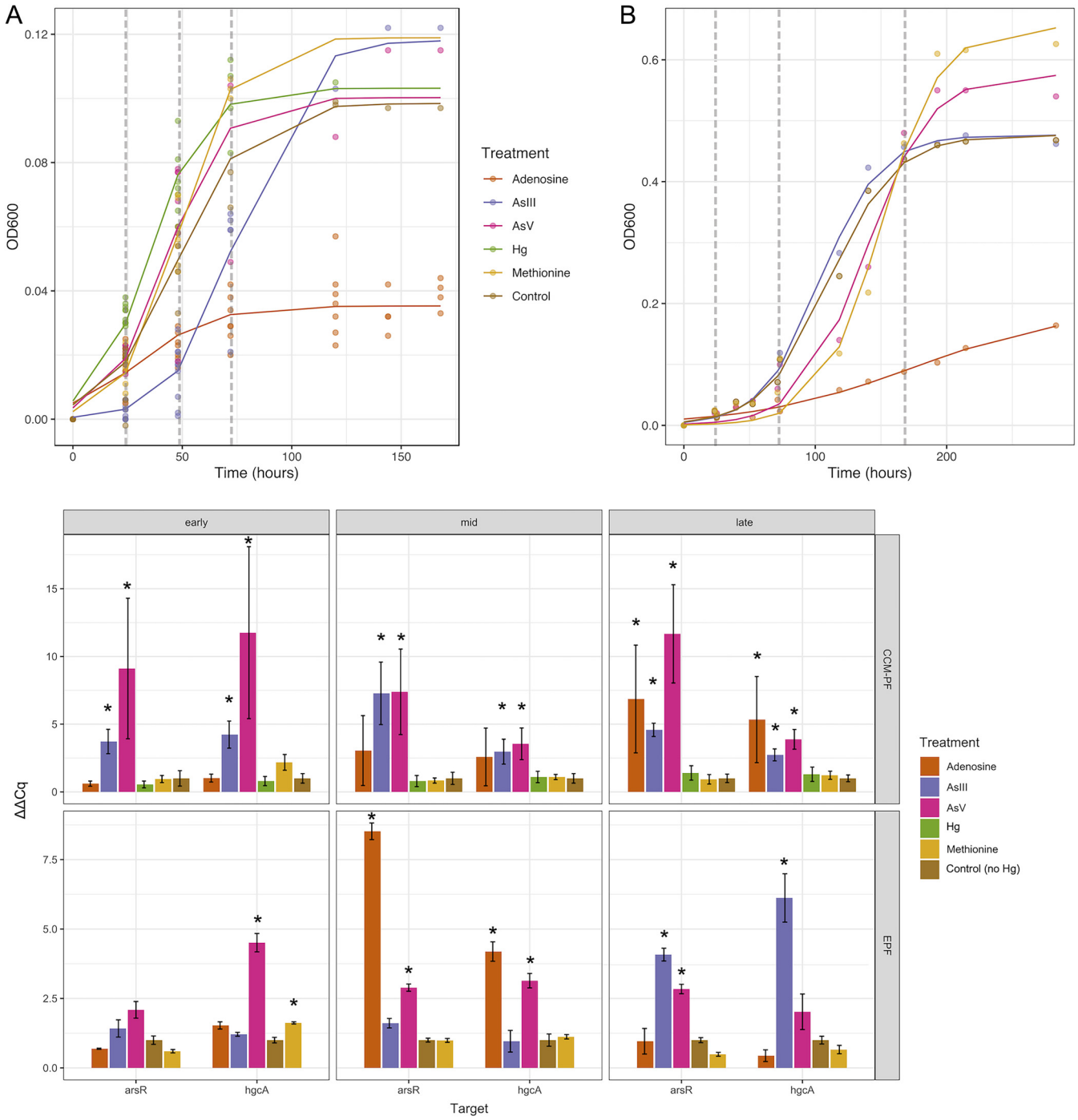


FIG 4 Regulation of *arsR* and *hgcA* expression in wild-type *Pseudodesulfovibrio mercurii* ND132. (Top) Growth curves for cells grown in defined CCM-PF (A) and enriched EPF (B) media with various potential transcriptional regulators. Growth plotted as change in optical density at 600-nm absorbance (OD_{600}) from inoculation at T_0, using the R program Growthcurver (97). Gray dashed lines indicate when cells were harvested for RNA extraction for qPCR at early, mid-, and late exponential growth. (Bottom) Relative expression ($\Delta\Delta C_q$) of *arsR2* (DND132_1054) and *hgcA* (DND132_1056) from qPCR of cDNA (from RNA). Expression of the target genes was normalized to three housekeeping genes (*gyrA*, *gyrB*, and *recA*) and to control (i.e., culture with no treatment). Treatments: 0.1 mM (CCM and EPF) adenosine-2′,3′-dialdehyde, 0.2 mM (CCM) and 0.02 mM sodium arsenite, 2.0 mM (CCM) and 0.2 mM (EPF) sodium arsenate, 1.0 mM HgCl (CCM only), and 1.0 mM (CCM and EPF) methionine. Asterisks indicate significant difference between $\Delta\Delta C_q$ of treatment and that of control (z-test, $P < 0.05$). qPCR was performed on biological triplicate cultures (CCM) and single culture (EPF) and included three analytical replicates per cDNA sample.

[Table S6]) of *arsR2* and *hgcA* in *P. mercurii* ND132 cultures than that in control when grown with an SAH-hydrolase blocker (i.e., adenosine-2′,3′-dialdehyde) and with As. The transcriptional response varied depending on culture conditions (defined versus enriched medium) and growth phase (i.e., early, mid-, or late exponential) (Fig. 4). In

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

defined CCM medium, we observed significantly higher expression (Table S6) of *arsR2* and *hgcA* in cultures grown with As(V) and As(III) across the growth curve (Fig. 3). We observed significantly higher expression of *arsR2* and *hgcA* from adenosine dialdehyde only at late exponential phase (Fig. 4). Relative expression of *arsR2* and *hgcA* in enriched medium differed from that under defined medium conditions. In enriched medium, we observed significantly higher *hgcA* expression in As(V)-amended cultures early in exponential growth and later in exponential growth for As(III)-amended cultures (Table S6). Compared to defined medium, we observed significantly higher *hgcA* expression in methionine- and adenosine-amended cultures at earlier stages of growth (Fig. 4). We attribute some of these differences to the presence of additional sulfur-containing metabolites in yeast extract (e.g., methionine) that could have contributed to increased SAH buildup in cultures in enriched medium when grown with methionine or adenosine dialdehyde. In defined medium (without yeast extract), the 1 mM methionine amendment did not induce a transcriptional response. Unlike under defined medium conditions, *arsR2* transcriptional changes did not mirror *hgcA* expression patterns.

Differences in *hgcA* expression observed across the growth curve could indicate that the transcriptional response is regulated by an unknown substrate or by an As species produced by *P. mercurii* ND132. This effect could vary over the growth curve, as the substrate or products of As metabolism build up in culture or the cell over time. Arsenic speciation analyses showed that *P. mercurii* ND132 reduces As(V) to As(III) and produces an unknown thioarsenic species from As(III) (Fig. S6). Differences in expression profiles between media could also be attributed to differences in As concentrations, with 10-fold less As(III) and As(V) used in enriched EPF medium than in defined CCM medium. Reducing the amount of starting As(III) from 0.2 mM to 0.02 mM could have increased the amount of time needed to reach a threshold for transcriptional response. The lower As(III) concentration used in EPF medium may explain why increased *arsR2* and *hgcA* expression was observed in As(III)-amended EPF cultures only at late exponential growth (Fig. 4). The higher starting concentration of As(V) (0.2 mM) than of As (III) (0.02 mM) in enriched EPF medium may account for the earlier observed transcriptional response in As(V)-amended cultures. Overall, these results show that growing *P. mercurii* ND132 with As(III) or As(V) will significantly increase *hgcA* expression.

Metabolites and medium components can confound expression analysis. When cells were washed of culture medium and then resuspended in basal salts with treatments, we observed significantly higher *arsR2* and *hgcA* expression due to As but not SAH (Fig. 5). Washing cells of medium components removed effects of adenosine dialdehyde on *arsR2* and *hgcA* expression observed in culture. Cells that were grown in enriched medium with the various treatments were harvested at mid-log phase (OD_{600} = 0.2) (Fig. 4B), kept anaerobic, washed of medium, and resuspended in a phosphate-salt buffer with the different treatments and 1 nM Hg. This was the same culture in which we measured *arsR2* and *hgcA* expression (Fig. 4, EPF mid-log panel). The difference in observed *arsR2* and *hgcA* expression in washed cells (Fig. 5C) from that in the culture (before washing) (Fig. 4) indicates that buildup of SAH (or another metabolite) during growth with methionine and adenosine dialdehyde likely causes the observed transcriptional response of *arsR2* (DND132_1054). The transcriptional response to As, as either As (III) or As(V), is more direct and instantaneous. We attribute differences in the transcriptional response to As over the growth curve as resulting from changes in As speciation and toxicity. The qPCR results from washed cells and batch cultures indicate that the *hgcA* regulator (DND132_1054) is responsive to both As and SAH.

As affects Hg methylation rates in ND132. We directly measured the impact of As and SAH on Hg methylation to help confirm that changes in *hgcAB* expression affected Hg methylation rates. Since *hgcA* transcription is regulated by *arsR2* in response to As and SAH, we expected As and SAH should also impact Hg methylation. Arsenic significantly affected *P. mercurii* ND132 Hg methylation rates, but effects varied depending on whether cultures were washed of medium components prior to methylation assays. We performed Hg methylation assays in both cultures and washed cells using mid-log

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

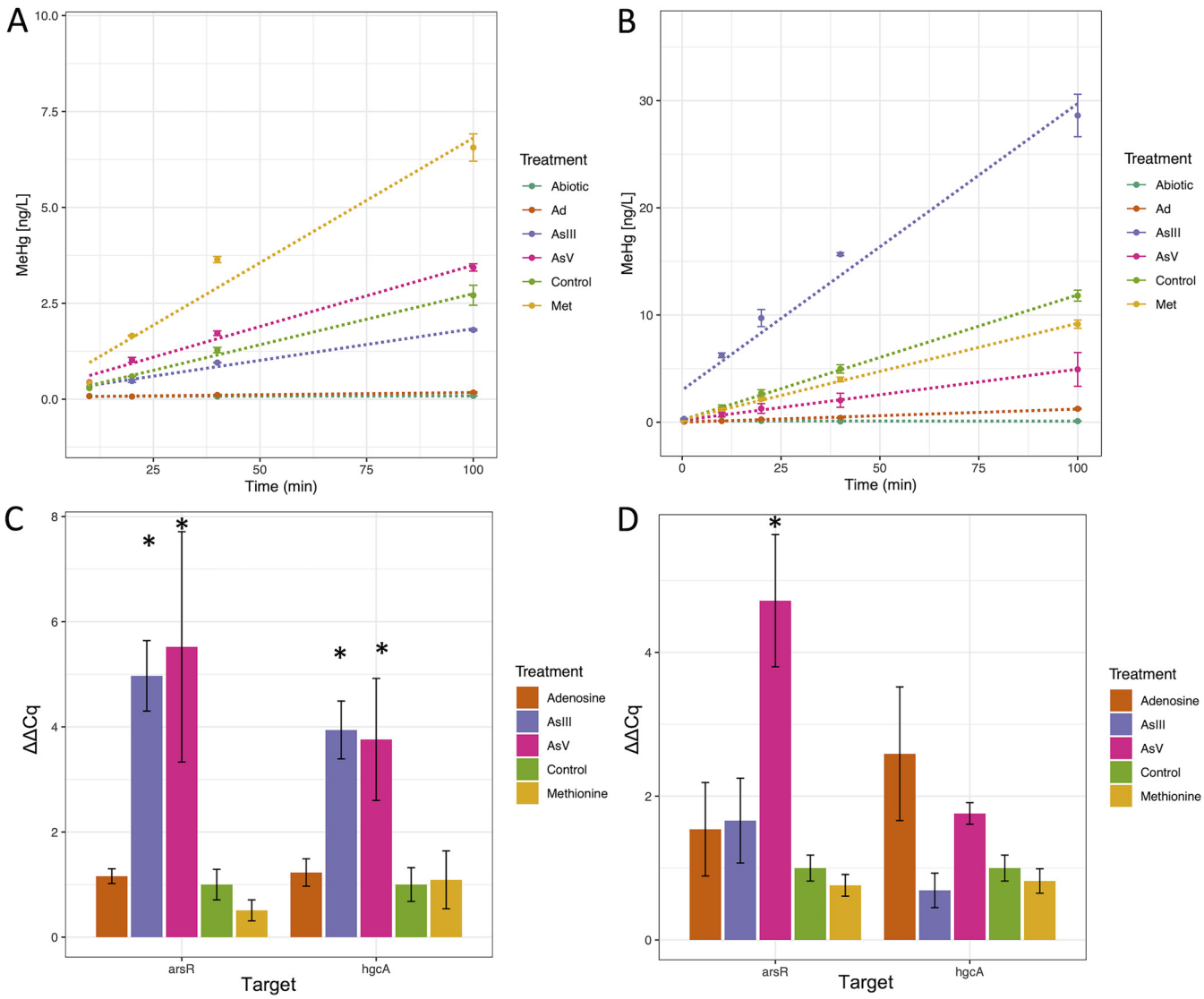


FIG 5 Impact of potential *hgcA* transcription regulators on MeHg production by *Pseudodesulfovibrio mercurii* ND132. (Top) MeHg production in washed-cell (A) and batch culture (B) assays with treatments: 0.1 mM adenosine-2′,3′-dialdehyde, 0.02 mM sodium arsenite, 0.2 mM sodium arsenate, 1.0 mM methionine, control (cells in unamended washed buffer or EPF medium), and abiotic treatment (no cells, unamended wash buffer or EPF medium). Linear regression fit to data using R. Hg methylation assays were performed in triplicates, with sampling at 0, 10, 20, 40, and 100 min after being spiked with 1 nM ^{201}HgCl. Error bars represent ±1 standard error of the mean. Cells were harvested from assays at t = 100 min for qPCR of cDNA (from RNA) to measure normalized expression ($\Delta\Delta C_q$) of *arsR* (DND132_1054) and *hgcA* (DND132_1056) in washed cells (C) and batch culture (D). Expression normalized to three housekeeping genes (*gyrA*, *gyrB*, and *recA*) and control (see Supplemental Methods in the supplemental material). Asterisks indicate significant difference between $\Delta\Delta C_q$ of treatment and that of control (z-test, $P < 0.05$).

cultures (OD_{600} of ~0.2, Fig. 4B) grown in EPF medium. Prior to methylation assays, cells were inoculated into medium amended with observed expression regulators of *hgcA*: methionine, adenosine dialdehyde, As(III), or As(V). In washed cells harvested at mid-log phase, methionine- and As(V)-treated cells produced significantly more MeHg than the control, while As(III) and adenosine significantly reduced MeHg production (Fig. 5A). In unwashed cultures, Hg methylation rates were significantly higher in As(III)-amended cultures, but significantly lower in cultures grown with As(V) or adenosine, than in control (Fig. 5B). In batch culture assays, there was no significant difference in MeHg produced in cultures grown with methionine compared with control.

Differences in metabolic activity and biomass could account for some of the differences in MeHg production between treatments. When grown with As(V) and methionine, *P. mercurii* ND132 had a steeper growth curve and reached a higher optical density than

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

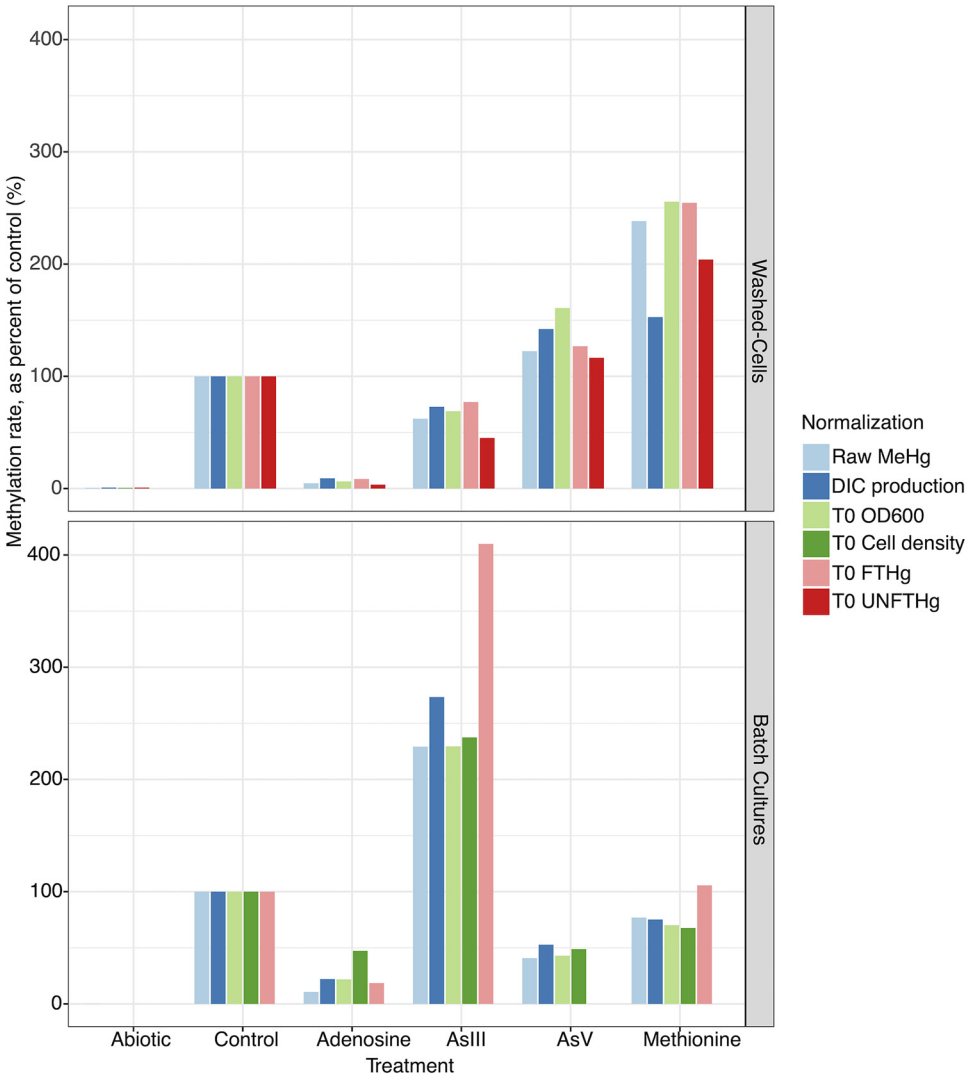


FIG 6 Normalized mercury methylation rates from culture and washed-cell assays. Mercury methylation rates in *Pseudodesulfovibrio mercurii* ND132 washed-cell (top panel) and batch culture (bottom panel) assays with test treatments compared to control when normalized to dissolved inorganic carbon (DIC) production (i.e., CO_2 respiration), optical density (OD_{600}) at T_0, cell density at T_0, filter-passable total Hg at T_0, and total unfiltered Hg at T_0.

it did with control and other treatments (Fig. 4B). In the washed-cell assays, the higher metabolic activity (CO_2 respiration rate) (Fig. S7) in the methionine treatment could explain some of the higher MeHg production (Fig. 5A). In As(V)-treated washed cells, higher MeHg production could not be explained by differences in metabolic activity, biomass, or unfiltered Hg at T_0 (Fig. 6 and Fig. S8). Normalization of methylation rates to various measures of cell growth and metabolism did not change the relative impact of treatments (Fig. 6). The measured initial concentration of total ^{201}Hg was the same in all bottles, and Hg mass balance was maintained in all bottles (Fig. S9). However, the filterable (0.22-μm filters) ^{201}Hg concentration varied among the treatments. Nevertheless, normalization of methylation rates to initial filterable ^{201}Hg concentrations did not alter the relative impact of treatments (Fig. 6).

In the Hg methylation assays performed with mid-log cultures, significantly more MeHg was produced in the As(III) treatment (Fig. 5B). This effect held true even when normalized to biomass (OD_{600}, cell counts) (Fig. S10), pH (Fig. S11), metabolic activity (Fig. S12), and total Hg (Fig. 5 and Fig. S13). Hg methylation rates in As(V)-amended cultures remained significantly lower than those in control when normalized for these

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

factors (Fig. 6). In washed cells and culture medium, adenosine-2′,3′-dialdehyde appears to limit cell growth and metabolic activity of *P. mercurii* ND132 (Fig. 4). MeHg production in adenosine-amended washed cells and batch culture assays was only slightly higher than that in abiotic controls. It was difficult to decouple the effects of lower metabolic activity and less biomass on limited MeHg production. However, normalizing for biomass (as cell density) put Hg methylation rates on par with those of As(V)-amended cultures in medium (Fig. 6).

Differences in Hg bioavailability may contribute to the differences in MeHg production among treatments. Sulfide has strong effects on Hg bioavailability for methylation (12, 30, 67, 68), and although grown without sulfate, ND132 cleaves the S−C bond in cysteine provided in growth medium to release small amounts of sulfide, generally below 10 μM (69). We consistently measured significantly less sulfide in filtered samples from As(V)-amended washed-cell (Fig. S7) and batch culture (Fig. S11) assays than in samples from other treatments, including As(III). This observation indicates that As (V) impacts sulfide speciation, possibly by disrupting metabolic pathways that produce reduced sulfide or precipitation of As species with sulfide. The higher concentrations of As(V) (0.2 mM) could account for the observed differences from As(III) (0.02 mM). In addition, the dissolved ^{201}Hg spike is rapidly lost to cell walls and any precipitates in the medium. We observed differences in filterable ^{201}Hg levels among treatments in both washed cells (Fig. S9) and batch cultures (Fig. S13). To test the impact of amendments on bioavailable Hg, we normalized methylation rates to the initial concentration of ^{201}Hg in each treatment. Because Me^{201}Hg is exported from cells after production (17), filterable total ^{201}Hg concentrations after T_0 often reflect Me^{201}Hg production (Fig. 6).

Hg methylation rates decoupled from *hgcA* transcription. Increased *hgcAB* transcriptional activity did not explain increased MeHg production in most treatments. With the exception of the As(V) washed cells, we did not observe significantly higher Hg methylation by cells with higher *hgcA* expression than that in control (Fig. 5). Despite seeing significantly higher MeHg production by washed cells amended with methionine and cultures grown with As(III), we did not observe significant differences in *hgcA* expression compared with control in qPCR analyses (Fig. 5). Washed cells amended with As(III) had significantly higher *hgcA* expression (Fig. 5) but produced significantly less MeHg (Fig. 6). Similarly, cultures grown with adenosine-2′,3′-dialdehyde and As(V) treatments had higher expression of *hgcA* but produced significantly less MeHg than control (Fig. 5). This finding was surprising as we expected increased *hgcA* expression to result in higher Hg methylation rates. However, *hgcA* activity may not have predicted higher Hg methylation rates because of several factors that we could not account for in this experiment: (i) increased transcription of *hgcAB* may not have resulted in increased translation of HgcAB, (ii) Hg availability and uptake by cells may be more rate-limiting than *hgcA*/HgcA activity, (iii) Hg complexation with As may interfere with Hg availability for methylation (70), and (iv) there may be possible competition between As and Hg for binding to HgcAB. Furthermore, the increased Hg methylation rate observed in As(III)-amended cultures could be from increased activity of other unknown proteins essential to the biochemical Hg methylation reaction.

As and adenosine increase *hgcA* and *hgcB* expression in ND132 transcriptomes. Transcriptome sequencing (RNA-seq) analyses of RNA extracted from mid-log washed cells and batch cultures confirm that both *hgcA* expression and *hgcB* expression are regulated by As and SAH. Growing *P. mercurii* ND132 with As or adenosine dialdehyde appeared to have a significant influence on the transcriptome (Fig. S14). We observed significantly higher expression (|log fold change [FC]| of $>$1.5, $P < 0.05$) of *arsR2* (DND132_1054), *hgcA* (DND132_1056), and *hgcB* (DND132_1057) in As(III)- and As(V)-treated washed cells than in control (Fig. 7). These features were also significantly differentially expressed ($P < 0.05$) in batch cultures amended with As(V) and adenosine dialdehyde (Fig. S15). The RNA-seq data show that *hgcB* is being upregulated with *hgcA*, which is important since both genes are needed for Hg methylation (19). Overall, there were more significant shifts in batch culture transcriptomes compared to those of washed cells, with many of the same features being differentially expressed across

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

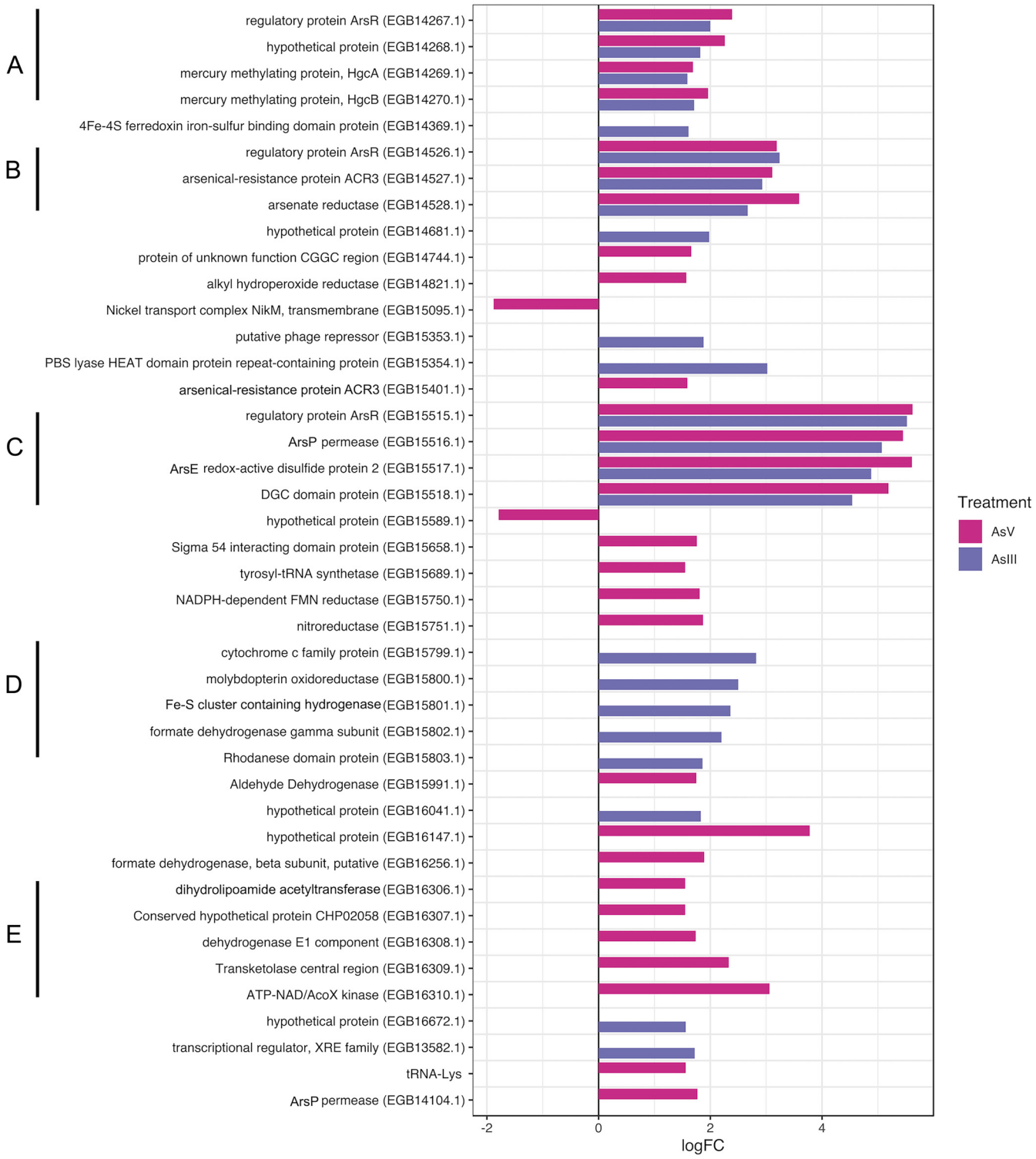


FIG 7 Differential expression of significant features (|log FC| of >1.5, $P < 0.05$) from RNA-seq analyses of As(V) (0.2 mM)- and As(III) (0.02 mM)-treated washed cells of wild-type *Pseudodesulfovibrio mercurii* ND132. Several putative operons (A to E) had significant differential expression compared to control washed cells (i.e., no arsenic treatment).

treatments (Fig. S14). Methionine-amended batch cultures had the least significant shift in their transcriptome compared to control ($n = 4$) (Fig. S16). In *P. mercurii* ND132 cultures grown with adenosine, the predicted SahR regulator encoded by *arsR4* (gene ID DND132_1481) was significantly upregulated (log FC = 2.36, $P = 2.64E-19$) along with the adenosylhomocysteinase-encoding gene *ahcY* (gene ID DND132_1480, log

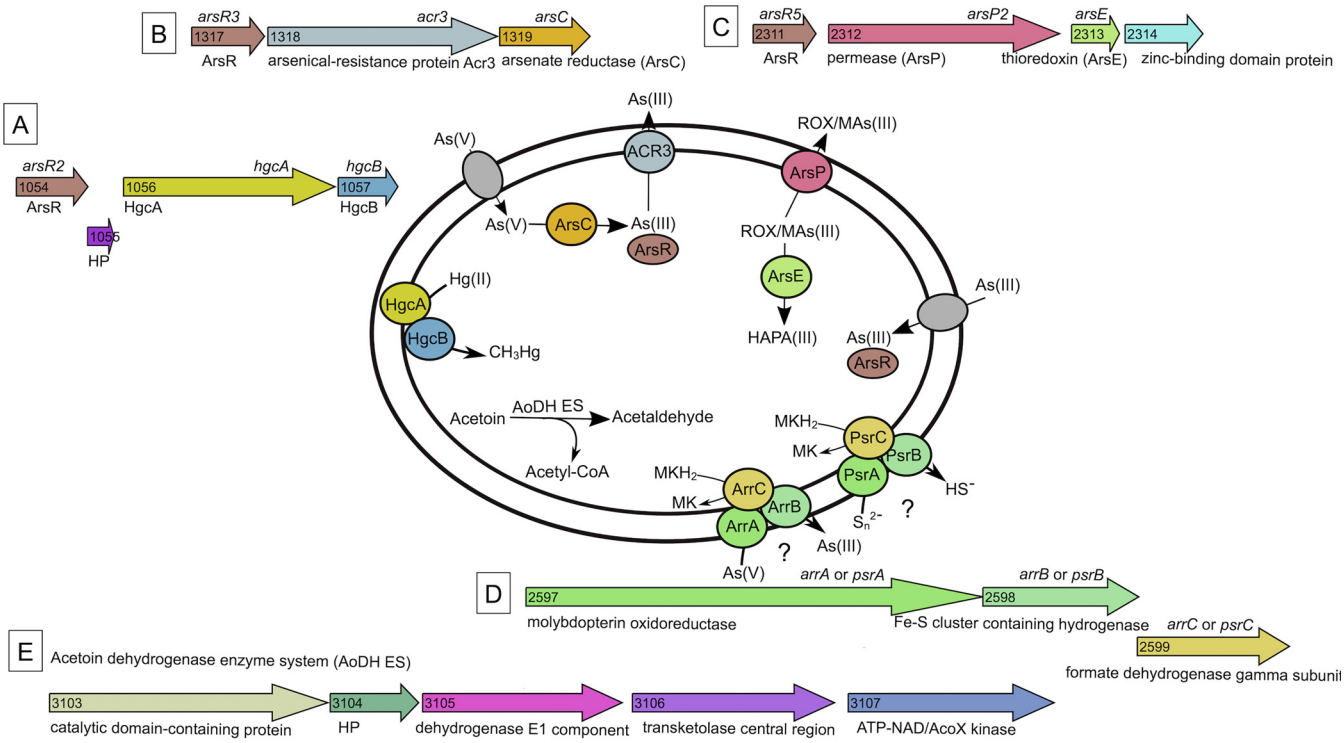


FIG 8 Proposed biochemical reactions for operons that were significantly differentially expressed in *Pseudodesulfovibrio mercurii* ND132 transcriptomes grown with arsenic. These include genes that are part of the ArsR regulon in ND132, including Hg methylation genes *hgcA* and *hgcB* (A), an operon that encodes arsenate reductase (ArsC) and As(III) efflux pump Acr3 (B), and an operon that includes homologues of organoarsenic resistance proteins ArsE and ArsP (C). Also significantly differentially expressed was an operon that encodes homologues of the polysulfide reductase (Psr) or arsenic respiration complex (Arr), which couples menaquinone (MK) oxidation to periplasmic As(V) or sulfur reduction (D), and a set of genes that encode homologues of the acetoin dehydrogenase enzyme system (AoDH ES) (E). *P. mercurii* ND132 is a confirmed Hg methylator and can reduce As(V) to As(III). The biochemical mechanism that produces the unidentified thioarsenic compound is unknown. To date, *P. mercurii* ND132 has not been tested for organoarsenic resistance or for acetoin degradation. The mechanism for uptake of As(V) and As(III) by ND132 is unknown. Operon graphic made using Gene Graphics (96).

FC = 2.31, P = 2.16E−16) (Fig. S16). The SAH-responsive regulator was not significantly upregulated in any other transcriptome, including methionine-treated cultures. This observation indicates that there may not have been enough buildup of SAH in *P. mercurii* ND132 cultures amended with 1 mM methionine to induce a transcriptional response from SahR or ArsR. There appears to be a cross-reactivity to SAH and As by two ArsR proteins encoded in the genome of *P. mercurii* ND132 *arsR2* (DND132_1054) as well as *arsR5* (DND132_2311) (Fig. S15). However, exposure to As(III) or As(V) did not induce a transcriptional response by SahR (*arsR4*, DND132_1481).

Coexpression of *hgcAB* with operons of arsenic resistance. *Pseudodesulfovibrio mercurii* ND132 is resistant to As(V) and As(III) and may also be resistant to organoarsenicals and respire As(V). Both Hg methylation genes (*hgcA* and *hgcB*) appear to be part of the ArsR regulon and are coregulated with *ars* operons. Based on RNA sequencing, there were 12 shared features (Fig. S14) that had significant differential expression (|log FC| of >1.5, P value of <0.05) in As-treated washed cells compared to control (Fig. 7). These features were also significantly differentially expressed ($P < 0.05$) in batch cultures amended with As(III), As(V), and adenosine dialdehyde (Fig. S15). In addition to increased *hgcA* and *hgcB* expression, two ArsR-regulated *ars* operons had significantly higher expression in As-amended washed cells and batch cultures (Fig. 7 and Fig. S15). Operon B (Fig. 8) encodes ArsR3, an Acr3 As(III) efflux permease, and arsenate reductase. Another set of upregulated genes (operon C) encodes ArsR5 and enzymes homologous to those utilized in organoarsenical resistance. Operon C was also upregulated in the transcriptome of adenosine dialdehyde cultures. The other ArsP (DND132_0889) and Acr3 (DND132_2196) permeases encoded outside an operon were not consistently upregulated with the *ars* operons (Fig. 7 and Fig. S15). The only

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

arsR gene of *P. mercurii* ND132 that was not significantly differentially expressed in any transcriptome was *arsR1* (DND132_0890).

In As(III) washed cells and As(V) and adenosine batch cultures, an operon with possible links to arsenic or polysulfide respiration was upregulated compared to control. The set of genes (operon D, Fig. 8) encodes possible homologues of the As(V) respiratory reductase (Arr) or polysulfide reductase (Psr) complexes. Since transcription of these genes was increased by As(III), we speculated that their encoded function may relate to As respiration. The genes encode homologues of ArrA (DND132_2597), ArrB (DND132_2598), and ArrC (DND132_2599) and are annotated in *P. mercurii* ND132 as a molybdopterin oxidase, an Fe-S cluster-containing hydrogenase, and formate dehydrogenase gamma subunit protein, respectively. The Arr complex is related to the dimethyl sulfoxide-molybdopterin (DMSO-MPT) reductase family, with Arr subunit A having a Mo/W-*bis*(pyranopterin guanosine dinucleotide) cofactor in its active site and a [4Fe-4S] center and Arr subunit B having four [4Fe-4S] clusters (71). A diverse range of microorganisms can respire As(V) for energy using ArrAB (72), including *Chrysiogenes arsenatis*, *Shewanella* strain ANA-3, *Alkaliphilus oremlandii*, and *Geobacter* sp. strain OR-1 (73, 74). In some microorganisms, ArrAB can function as a bidirectional enzyme (75), oxidizing As(III) to As(V). Often, ArrAB are encoded in microbial genomes alongside genes of the *ars* operon. However, in *P. mercurii* ND132 the putative operons are not colocalized. The ND132 ArrA homologue includes the conserved molybdenum and Fe-S binding domains found in known functional respiratory As (V) reductases but is missing some of the key active-site residues (75).

Instead, operon D (Fig. 8) may encode polysulfide or thiosulfate reduction rather than As(V) respiration. Based on the phylogenetic relationship of the putative ArrA in *P. mercurii* ND132 with other DMSO-MPT family proteins, the protein encoded by DND132_2597 is more closely related to bacterial polysulfide reductase (PsrA) than ArrA from known As(V)-respiring bacteria (Fig. S17). Microbes in deep-sea vents and hydrothermal springs use the membrane protein complex PsrABC to couple periplasmic polysulfide or thiosulfate reduction with quinone oxidation (76). The operon in *P. mercurii* ND132 also encodes homologues of cytochrome *c* oxidase (DND132_2596) and two rhodanese-domain proteins (DND132_2600 and DND132_2601), one of which (DND132_2600) is significantly upregulated in As(III) cells. A second cluster of putative *psrAB* genes (DND132_1314 and DND132_1313) is present elsewhere in the genome (Fig. S17) and likely encodes cytoplasmic sulfur reduction (77). The transcription of these *psrAB* genes was not significantly affected by As or SAH. The DMSO-MPT family of proteins that includes Psr and Arr has been implicated in thioarsenic production. The geothermal isolate *Pyrobaculum yellowstonensis* strain WP30 produces thioarsenate during respiratory elemental sulfur and As(V) reduction (78). More work is needed to know if the protein complex encoded by operon D interacts with As(III), As(V), or thiosulfates/polysulfides and enables *P. mercurii* ND132 to respire As. However, it may be responsible for producing the unknown thioarsenic species observed in *P. mercurii* ND132 culture medium.

In As(V) washed cells an operon with homology to the acetoin dehydrogenase complex was upregulated. In batch cultures, this operon was significantly downregulated in all treatments except methionine (Fig. S15). The operon is predicted to encode an acetoin dehydrogenase enzyme system (AoDH ES) that catalyzes the conversion of acetoin to acetyl-CoA and acetaldehyde (Fig. 8). The AoDH ES contains multiple copies of three enzymatic components: acetoin oxidoreductase (E1), dihydrolipoamide acetyltransferase (E2), and dihydrolipoyl dehydrogenase (E3) (79, 80). The operon (operon E, Fig. 8) encodes AoDH ES homologues annotated as a dihydrolipoamide acetyltransferase (DND132_3103) (Pfam PF00198, PF00364, and PF02817), acetoin:2,6-dichlorophenolindophenol oxidoreductase subunit alpha (DND132_3105) (dehydrogenase E1 component, Pfam PF00676), acetoin:2,6-dichlorophenolindophenol oxidoreductase subunit beta (DND132_3106) (transketolase central region, Pfam PF02779 and PF02780), and an acetoin catabolism protein with ATP-NAD/AcoX kinase motifs (Pfam PF01513). These proteins also share homology to enzymes in the pyruvate dehydrogenase complex that convert pyruvate into acetyl-CoA by

pyruvate decarboxylation (81). The operon is possibly regulated by a sigma-54-dependent HTH transcriptional regulator (DND132_3102). In closely related *Pseudodesulfovibrio hydrargyri* BerOc1, this protein is annotated as acetoin dehydrogenase operon transcriptional activator AcoR (UniProt accession no. A0A1J5NC98). No obvious link to As metabolism or resistance could be found. It is possible that this putative AoDH ES operon is upregulated in *P. mercurii* ND132 As(V) washed cells because of cross-reactivity of As with the HTH regulator (DND132_3102) or because of a knock-on effect from other differentially regulated mechanisms.

Summary. The broad research aim of these experiments was to examine transcriptional regulation of *hgcAB* in *Pseudodesulfovibrio mercurii* ND132, with a focus on the potential role of an upstream ArsR-like regulator found in *hgc*$^+$ *Deltaproteobacteria*. Additionally, we examined the relationship between Hg methylation rates in *Pseudodesulfovibrio mercurii* ND132 with metabolic activity and *hgcAB* transcriptional activity. We have shown that *hgcAB* expression in *Pseudodesulfovibrio mercurii* ND132 is transcriptionally regulated by an ArsR-like regulator encoded by *arsR2* (DND132_1054) that is responsive to As and SAH. Since *P. mercurii* ND132 can reduce As(V) to As(III), and we observed increased *hgcA* and *hgcB* expression when it was grown with both species, it is likely that As(III) coordinates with ArsR2. The *hgcAB* transcriptional regulator shares homology to other members of the ArsR/SmtB family but has a distinct four-cysteine motif in its metal(loid) binding site. This homology also distinguishes it from other ArsR family regulators encoded in the genome of *P. mercurii* ND132. Since As(III) preferentially binds to three cysteine residues in ArsR repressors of the *ars* operon, we cannot exclude the possibility that another As species or heavy metal(loid) cation may coordinate with ArsR2 and also control *hgcAB* expression. Although the transcriptional response to As was more pronounced, under some culture conditions methionine and adenosine dialdehyde also significantly increased *arsR2* and *hgcAB* transcription. We attribute this transcriptional response to a buildup in SAH within the cell.

The RNA-seq analyses show that *hgcAB* transcription is coregulated as part of the ArsR regulon with operons that encode As resistance. We also observed significant coexpression of *hgcAB* with genes that encode putative As(V) or polysulfide respiration when *P. mercurii* ND132 is grown with As. These pathways may explain the possible thioarsenic production observed in *P. mercurii* ND132 cultures. This indicates that As biogeochemistry may influence *hgcAB* expression and Hg methylator activity in some environmental settings. In this study, we did not observe a significant loss in As resistance with a loss in *hgcAB* function in *Pseudodesulfovibrio mercurii* ND132. However, the coexpression and colocalization of *hgcAB* with arsenic resistance genes in some *hgc*$^+$ genomes may hold clues to *hgcAB* evolution and possible environmental drivers of horizontal transfer of these genes. More work is needed to tease apart possible overlaps between HgcAB function and arsenic resistance and respiration mechanisms.

Results from these experiments do not confirm that higher *hgcAB* expression alone increases MeHg production but indicate that growth and culture conditions and the nature of *hgcAB* transcriptional regulation probably all play roles in methylation control. Both As(III) and As(V) impacted Hg methylation and transcription of *hgcAB* and *arsR* in *P. mercurii* ND132, although the impacts were not always coupled. We also observed that metabolic activity and microbial biomass can significantly influence Hg methylation rates in cultures and counter possible impacts of *hgcAB* activity. Hg methylation rates may also be impacted by HgcAB translation, protein folding, cofactor and membrane insertion, and the methyl donor mechanism. Combining information on *hgcAB* transcription with microbiome composition, metabolic activity, and Hg speciation and geochemistry may help resolve environmental Hg methylation rates. It is therefore important to have accurate and reliable methods for measuring these factors and using multiple parameters when incorporating microbial dynamics into predictive models for Hg methylation.

MATERIALS AND METHODS

Bacterial strains and culture media. For this study, we used the sulfate-reducing Hg methylator and incomplete oxidizer *Pseudodesulfovibrio mercurii* ND132 (previously *Desulfovibrio desulfuricans*

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

ND132) as a model organism for understanding the mechanism and genetic regulation of Hg methylation (9, 30). *P. mercurii* ND132 has a well-annotated genome and can grow on a number of substrates including lactate-sulfate, pyruvate-fumarate, formate-acetate-sulfate, and formate-acetate-fumarate (82). Furthermore, knockout strains (Δ*hgcA*,*B*) of *P. mercurii* ND132 have been used to show the requirement of *hgcA* and *hgcB* for Hg methylation (2, 19). For this study, we used *P. mercurii* ND132 wild-type and *hgcAB* deletion mutant cultures from Dwayne A. Elias' culture collection. Since these experiments began, wild-type *Pseudodesulfovibrio mercurii* ND132 has been deposited into the DSMZ (type strain DSM 110689) and ATCC (ATCC TSD-224) culture collections. The Hg methylation knockout strain (Δ*hgcAB*) of *P. mercurii* ND132 was originally sourced from Judy Wall (2). Stock cultures were stored at −80°C in lactate-sulfate medium and 10% (vol/vol) glycerol.

ND132 is capable of fumarate respiration using pyruvate as an electron donor (9). For most assays, *P. mercurii* ND132 was grown in this way on either CCM (83) or estuarine pyruvate fumarate (EPF) medium (9). Both media contain 40 mM pyruvate and 40 mM fumarate. CCM is a defined medium, while EPF medium includes 0.5% yeast extract. Both media were buffered to pH 7.2 using morpholinepropanesulfonic acid (MOPS) and include vitamin and mineral additions. Media were reduced with 0.1 mM (CCM) or 0.05 mM (EPF) cysteine-HCl to remove trace oxygen contamination. Other amendments varied by experiment. All media contained resazurin as a redox indicator. For some arsenic toxicity assays, ND132 was also grown on CCM medium with lactate (60 mM) and sulfate (30 mM) as electron donor and acceptor. All culture work was done at 31 ± 1°C.

Experimental assays for testing regulators of Hg methylation and *hgcAB* transcription. We evaluated the potential transcriptional response of a putative *hgcAB* regulator to *S*-adenosylhomocysteine (SAH) and arsenic. We also evaluated the potential influence of *hgcA* expression on Hg methylation using batch culture and washed-cell approaches. For all assays, cultures were started from freezer stocks. In all cases, cells were grown on CCM or EPF medium. Out of the freezer, cells were grown to mid- to late exponential phase and then inoculated at 10% (vol/vol) into medium containing one of the four test conditions or unamended control medium. In defined CCM medium the test amendments were 0.2 mM sodium arsenite, 2.0 mM sodium arsenate, 1 mM methionine, 0.1 mM adenosine-2′,3′-dialdehyde, and 1 nM HgCl. In EPF medium, arsenic concentrations were reduced to 0.02 mM sodium arsenite and 0.2 mM sodium arsenate. Amendment concentrations for expression and methylation studies were chosen based on toxicity tests (see below). Arsenic concentrations selected were the highest levels that allowed unimpacted growth, which was specific to each medium. In qPCR expression experiments conducted over the growth curve, both control and amendment cultures were grown without added Hg. In the washed-cell assays and Hg methylation assays performed in culture, all treatments including the control were spiked with 1 nM isotopically enriched ^{201}Hg.

Methylation assays were carried out in triplicate by the addition of 1 nM isotopically enriched ^{201}Hg (98.1% purity; Oak Ridge National Laboratory) (in 1% [vol/vol] HCl). Uninoculated sterile medium with and without treatment amendments served as abiotic controls. Methylation assays ran 100 min, after which aliquots of unfiltered culture were immediately acidified (0.5% [vol/vol] HCl) and frozen (−20°C) for Me^{201}Hg analysis. Experimental timeframes were selected on the basis of previous kinetic experiments demonstrating a plateau in MeHg production by strain ND132 (17).

Methylation assays were performed using either washed cells or batch culture cells. For washed-cell assays, cells were harvested at mid-log phase (OD_{600} of ~0.2) from batch culture and washed and resuspended in a minimal wash buffer as described previously (69). The salt buffer contained 1 mM pyruvate and 1 mM fumarate, reduced with 0.5 mM cysteine-HCl and buffered to pH 7.5. For batch culture experiments, culture aliquots from mid-log phase (OD_{600} of ~0.2) were directly spiked with ^{201}Hg. In most cases, the concentration of filterable and total ^{201}Hg was measured immediately after the ^{201}Hg spike and at the end of the methylation assays (and in some cases through time during the assays). Samples were filtered through 13-mm-diameter 0.22-μm CellTreat MCE membrane syringe filters. For most studies we also measured pH, optical density at 600-nm wavelength (OD_{600}), acridine orange direct cell counts (AODC), and sulfide concentration over the course of the assay. Additionally, for some studies we measured short-chain fatty acids and/or CO_2 production as indicators of cellular activity. All culture manipulations were carried out in a Coy glove bag under an O_2-free atmosphere.

Culture aliquots were saved for RNA extraction before the assays and from each assay replicate at the end of the time-course (t = 100 min). For RNA extraction, cells were harvested from batch cultures and methylation assays at 2,800 × g for 10 min. The cell pellet was resuspended in 1 mL TRIzol reagent (Invitrogen) and kept at −20°C prior to nucleic acid extraction.

Hg and MeHg analyses. Isotope-specific total Hg and MeHg concentrations were measured using modifications of EPA methods 1631 (EPA-821-R-01-023, March 2001) and 1630 (EPA-821- R-01-020, January 2001) respectively, with inductively coupled plasma mass spectrometry (ICP-MS) detection. Details of the analytical method and quality assurance (QA)/quality control (QC) are provided in Table S1 in the supplemental material.

Sulfide, organic acid, cell count, and gas analytical methods. Sulfide was analyzed using a sulfide selective electrode with an Ag−AgCl reference electrode (Thermo Scientific) and was calibrated against Pb-titrated Na_2S standards made in sulfide antioxidant buffer (SAOB). Saturated Na_2S stocks were made from washed crystals in degassed ultrapure water and held in N_2-filled sealed serum vials until use. SAOB contains 2 M NaOH, 0.2 M EDTA, and 0.2 M ascorbic acid in degassed ultrapure water and was sealed under N_2 until use. Organic acids (acetate, pyruvate, fumarate, and succinate) (0.5 to 200 μM) were determined with a Dionex Integrion high-performance ion chromatograph (HPIC) equipped with an AS11-HC column at 35°C with a KOH gradient of 1 to 60 mM with an ICS-5000$^+$ EG effluent generator

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

system (Thermo Fisher Scientific). Sample volumes of 10 μL were injected with a Dionex AS-AP autosampler at room temperature.

Cell counts were done manually using acridine orange staining. Cells were fixed at a dilution of 1:100, with 0.1 mL of cell culture preserved in 9.1 mL of 20 mM phosphate buffer solution (PBS) and 0.8 mL glutaraldehyde, and stored in the refrigerator prior to slide preparation. Slides were made by combining 0.2 mL of 0.1% acridine orange and 1.9 mL of fixed cells in microcentrifuge tubes and allowing 1 h at room temperature in the dark for staining. The stained cells were mixed with 1 mL of 20 mM PBS and collected on filters via vacuum. Counts were performed manually and averaged from 10 different grids over a 0.01-mm^2 area.

CO_2 production was measured over time in the headspace of serum vials containing culture aliquots, after purging the headspace with O_2-free N_2 for ~5 min. All measurements were made from duplicate aliquots for each treatment. The 100-mL serum vials contained either 5 mL of culture or 25 mL of washed cells in salt buffer. Headspace CO_2 was measured hourly after an initial equilibration period using a Li-Cor LI-7000 CO_2 analyzer. Total inorganic carbon production was calculated from headspace CO_2 concentrations based on temperature, pH, the Henry's Law constant for CO_2, and equilibrium among dissolved inorganic carbon (DIC) species. DIC production rates were calculated from the linear portion of the slope with time.

RNA extraction. Cell cultures were pelleted at 2,800 $\times$ *g* for 10 min, excess medium was removed, and then cultures were resuspended in 1 mL of TRIzol reagent (Invitrogen) and stored at −20°C until processing. RNA was extracted following the TRIzol protocol, which involves a phenol-chloroform extraction, isopropanol precipitation, and ethanol cleaning step. Extracted RNA was further purified using the RNeasy Clean-up kit (Qiagen) and DNase treatment with a Turbo DNA-free kit (Thermo Fisher). RNA concentration was measured using a Qubit RNA HS assay kit on the Qubit 3.0 fluorometer (Thermo Fisher), and purity was determined by NanoDrop (Thermo Fisher). Genomic DNA (gDNA) contamination was assessed by performing a qPCR with DNase-treated RNA prior to reverse transcription. cDNA synthesis was performed using the iScript cDNA synthesis kit (Bio-Rad), and cDNA concentration was measured using a Qubit single-stranded DNA (ssDNA) assay kit on the Qubit 3.0 fluorometer (Thermo Fisher). cDNA was diluted to 1 ng/μL with RNase-free water prior to qPCR.

qPCR. Real-time quantitative PCR (qPCR) mixtures were set up on a 96-well plate with each standard and biological cDNA sample run in triplicate. Each well contained 1 ng of cDNA, 0.25 μM forward and reverse primer, 0.02 μL ROX reference dye (Bio-Rad), 10 μL iQ SYBR Green Supermix (Bio-Rad), and 1$\times$ Tris-EDTA (TE) buffer (pH 8), filled to a 20-μL total volume. For standard curves, we used serial dilutions of 4.2 ng/μL (measured on Qubit) gDNA extracted from pure culture of *P. mercurii* ND132 wild type, estimated to be equivalent to 10^6 copies/μL based on the size of the genome (3.8 Mbp) (82). All standards and samples of each primer set were amplified on the same plate, with each plate containing only one target or reference gene. All qPCR analyses from the same experiment were performed in succession with the same standard curve. Custom-designed oligonucleotide primers were used to amplify a 107-base pair (bp) target region of *hgcA* (DND132_1056- hgcA-F and DND132_1056-hgcA-R) and a 180 bp region of the *arsR*-like gene before *hgcA* (DND132_1054-arsR-F and DND132_1054-arsR-R) (Table S2). Three single-copy housekeeping genes that encode recombinase subunit A (*recA*), gyrase subunit A (*gyrA*), and gyrase subunit B (*gyrB*) were used for normalization of expression of genes of interest. These housekeeping genes were chosen for normalization due to their observed expression stability in ND132 wild-type transcriptomes (data not shown). Custom primers were designed to amplify a 153 bp region of *recA* (DND132_1603, DND132_1603-recA-F and DND132_1603-recA-R), a 148 bp region of *gyrA* (ND132_2285, DND132_2285-gyrA-F and DND132_2285-gyrA-R), and a 182 bp region of *gyrB* (DND132_2284, DND132_2284-gyrB-F and DND132_2284-gyrB-R). Primers were designed to function over the same annealing temperature (65°C), and the same thermocycler protocol was used to amplify each gene: polymerase activation for 3 min at 95°C, followed by 32 cycles of denaturing for 15 s at 95°C and annealing for 20 s at 65°C, followed by a melt curve of 0.1°C/s from 65°C to 95°C, ending with 15 s at 95°C.

RNA-seq transcriptomics. RNA sequencing was performed on clean, DNase-treated RNA extracted from the batch culture and washed-cell Hg methylation assays. For the washed-cell assays, only RNAs from the control, As(III), and As(V) treatments were sent for RNA-seq. All treatments from the culture Hg methylation assays were sent for sequencing. Three biological replicates for each treatment were sequenced. Library preparation and sequencing were performed by Genewiz (Azenta Life Sciences, NJ). rRNA depletion was performed during library preparation using three probes from the Qiagen FastSelect rRNA 5S/16S/23S kit (Qiagen, Hilden, Germany), respectively. RNA sequencing libraries were prepared using the NEBNext Ultra II RNA library preparation kit for Illumina by following the manufacturer's recommendations (New England Biolabs [NEB], Ipswich, MA, USA). Briefly, enriched RNAs are fragmented for 15 min at 94°C and then first- and second-strand cDNAs are synthesized. cDNA fragments are end repaired and adenylated at the 3′ ends, and universal adapters are ligated to cDNA fragments, followed by index addition and library enrichment with limited-cycle PCR. Sequencing libraries were validated using the Agilent TapeStation 4200 (Agilent Technologies, Palo Alto, CA, USA) and quantified using a Qubit 2.0 fluorometer (Thermo Fisher Scientific, Waltham, MA, USA) as well as by quantitative PCR (Kapa Biosystems, Wilmington, MA, USA). Multiplexed and clustered sequencing libraries were sequenced using Illumina HiSeq 2- by 150-bp paired-end configuration.

Raw sequencing files were uploaded to KBase (84). Initial read quality was assessed with FastQC, followed by adapter and quality-based trimming with Trimmomatic (85), alignment to the *Pseudodesulfovibrio mercurii* ND132 reference genome (https://www.ncbi.nlm.nih.gov/assembly/GCF_000189295.2) with HISAT2 (86), and assembly of transcripts from alignment using StringTie (87). Differential expression analysis was performed with DESeq2 (88), the results were uploaded to the Galaxy web platform (89), and we used the public server at usegalaxy.org to produce figures using the Galaxy R scripts. The paired-end RNA sequencing files for the washed-cell (90) and batch culture (91) assays are publicly available through Figshare.

Arsenic analyses. Arsenic speciation analyses were done at Brooks Applied Labs (Bothell, WA, USA). Samples were collected from culture bottles, filtered (Whatman cellulose acetate membrane syringe filters, 0.45 μm) into BD Vacutainers, preserved in EDTA, and stored at 4°C in the dark until analyzed. Arsenic (As) speciation analysis, including arsenite [As(III)], arsenate [As(V)], monomethylarsonic acid (MMAs), dimethylarsinic acid (DMAs), and unknown As species were determined on all samples. Arsenic species were chromatographically separated on an ion exchange column and then quantified using inductively coupled plasma collision reaction cell mass spectrometry (ICP-CRC-MS). The results were not method blank corrected and were evaluated using reporting limits adjusted to account for sample aliquot size.

Structural modeling. Structural models of the putative ArsR-like transcriptional regulator from *P. mercurii* ND132 (DND132_1054, UniProt ID F0JBE8) were generated with ColabFold (61). ColabFold is an implementation of AlphaFold2 (62) that uses MMseqs2 (92) to generate large multiple-sequence alignments by searching the UniRef and MGnify databases (93). Based on its sequence similarity to other transcriptional regulators that are known to function as homodimers, we modeled the homodimeric form of the protein using AlphaFold-Multimer (94), which is a version of AlphaFold2 trained specifically to model protein complexes. Models were ranked by their average predicted local distance difference test (pLDDT) and predicted TM score (pTMscore), and the model with the highest pTMscore was selected for further analysis.

Statistical methods. Statistical modeling was performed in JMP, version 16 (SAS Institute Inc., Cary, NC, 1989 to 2022), and R statistical software (version 4.2.0) (95). Generalized linear models were used to evaluate differences in Hg methylation rates, inorganic carbon production, organic acid production, and growth rates among treatments. Where appropriate, models included time, treatment, and any potential interaction terms. In the absence of significant interactions, differences among treatments were evaluated using pairwise comparison of slopes. For qPCR results, a two-sample z-test was used to determine if the difference in *arsR* and *hgcA* expression was significant between treatments and control. For significance testing, the mean and standard deviation for quantification cycle ($\Delta\Delta C_q$) values for each target gene and treatment were compared to control.

SUPPLEMENTAL MATERIAL

Supplemental material is available online only.

SUPPLEMENTAL FILE 1, PDF file, 9.2 MB.

ACKNOWLEDGMENTS

Caitlin M. Gionfriddo was a Robert and Arlene Kogod Secretarial Scholar with the Smithsonian Environmental Research Center while conducting work described in this paper. This work was funded by the U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research, Subsurface Biogeochemical Research (SBR) Program, and is a product of the Critical Interfaces Science Focus Area at Oak Ridge National Laboratory, which is managed by UT-Battelle, LLC, under contract no. DE-AC05-00OR22725 with the U.S. Department of Energy.

The As speciation analysis was performed by Brooks Applied Labs, and we thank Ben Wozniak and Collette Machado for technical support. The RNA-seq transcriptome analyses were performed at Azenta US Inc., and we thank Andrew Corbin and Eric Danzeisen for technical support. We thank the reviewers for taking the time and effort to review the manuscript and for their insightful comments and suggestions.

REFERENCES

1. Branfireun BA, Cosio C, Poulain AJ, Riise G, Bravo AG. 2020. Mercury cycling in freshwater systems-an updated conceptual model. Sci Total Environ 745:140906. https://doi.org/10.1016/j.scitotenv.2020.140906.
2. Parks JM, Johs A, Podar M, Bridou R, Hurt RA, Smith SD, Tomanicek SJ, Qian Y, Brown SD, Brandt CC, Palumbo AV, Smith JC, Wall JD, Elias DA, Liang L. 2013. The genetic basis for bacterial mercury methylation. Science 339:1332–1335. https://doi.org/10.1126/science.1230667.
3. Podar M, Gilmour CC, Brandt CC, Soren A, Brown SD, Crable BR, Palumbo AV, Somenahally AC, Elias DA. 2015. Global prevalence and distribution of genes and microorganisms involved in mercury methylation. Sci Adv 1: e1500675. https://doi.org/10.1126/sciadv.1500675.
4. McDaniel EA, Peterson B, Stevens SL, Tran PQ, Anantharaman K, McMahon K. 2020. Expanded phylogenetic diversity and metabolic flexibility of microbial mercury methylation. mSystems 5:e00299-20. https://doi.org/10.1128/mSystems.00299-20.
5. Gilmour CC, Podar M, Bullock AL, Graham AM, Brown SD, Somenahally AC, Johs A, Hurt RA, Jr, Bailey KL, Elias DA. 2013. Mercury methylation by novel microorganisms from new environments. Environ Sci Technol 47: 11810–11820. https://doi.org/10.1021/es403075t.
6. Gionfriddo CM, Wymore AM, Jones DS, Wilpiszeski RL, Lynes MM, Christensen GA, Soren A, Gilmour CC, Podar M, Elias DA. 2020. An improved *hgcAB* primer set and direct high-throughput sequencing expand Hg-methylator diversity in nature. Front Microbiol 11:541554. https://doi.org/10.3389/fmicb.2020.541554.
7. Capo E, Peterson BD, Kim M, Jones DS, Acinas SG, Amyot M, Bertilsson S, Björn E, Buck M, Cosio C, Elias DA, Gilmour C, Goñi-Urriza M, Gu B, Lin H, Liu YR, McMahon K, Moreau JW, Pinhassi J, Podar M, Puente-Sánchez F, Sánchez P, Storck V, Tada Y, Vigneron A, Walsh DA, Vandewalle-Capo M, Bravo AG, Gionfriddo CM. 2023. A consensus protocol for the recovery of mercury methylation genes from metagenomes. Mol Ecol Resour 23: 190–204. https://doi.org/10.1111/1755-0998.13687.
8. Goñi-Urriza M, Klopp C, Ranchou-Peyruse M, Ranchou-Peyruse A, Monperrus M, Khalfaoui-Hassani B, Guyoneaud R. 2020. Genome insights of mercury methylation among Desulfovibrio and Pseudodesulfovibrio strains. Res Microbiol 171:3–12. https://doi.org/10.1016/j.resmic.2019.10.003.
9. Gilmour CC, Soren AB, Gionfriddo CM, Podar M, Wall JD, Brown SD, Michener JK, Urriza MSG, Elias DA. 2019. Pseudodesulfovibrio mercurii sp. nov., a mercury-methylating bacterium isolated from sediment. Int J Syst Evol Microbiol 71:e004697. https://doi.org/10.1099/ijsem.0.004697.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

10. Ranchou-Peyruse M, Goñi-Urriza M, Guignard M, Goas M, Ranchou-Peyruse A, Guyoneaud R. 2018. Pseudodesulfovibrio hydrargyri sp. nov., a mercury-methylating bacterium isolated from a brackish sediment. Int J Syst Evol Microbiol 68:1461–1466. https://doi.org/10.1099/ijsem.0.002173.
11. Gionfriddo C, Capo E, Peterson B, Lin H, Jones D, Bravo A, Bertilsson S, Moreau J, McMahon K, Elias D, Gilmour C. 2021. Hg-MATE-Db v1.01142021. https://doi.org/10.25573/serc.13105370.
12. Gilmour CC, Bullock AL, McBurney A, Podar M, Elias DA. 2018. Robust mercury methylation across diverse methanogenic archaea. mBio 9:e02403-17. https://doi.org/10.1128/mBio.02403-17.
13. King JK, Kostka JE, Frischer ME, Saunders FM. 2000. Sulfate-reducing bacteria methylate mercury at variable rates in pure culture and in marine sediments. Appl Environ Microbiol 66:2430–2437. https://doi.org/10.1128/AEM.66.6.2430-2437.2000.
14. King J, Kostka J, Frischer M, Saunders FM, Jahnke R. 2001. A quantitative relationship that demonstrates mercury methylation rates in marine sediments are based on the community composition and activity of sulfate-reducing bacteria. Environ Sci Technol 35:2491–2496. https://doi.org/10.1021/es001813q.
15. Date SS, Parks JM, Rush KW, Wall JD, Ragsdale SW, Johs A. 2019. Kinetics of enzymatic mercury methylation at nanomolar concentrations catalyzed by HgcAB. Appl Environ Microbiol 85:e00438-19. https://doi.org/10.1128/AEM.00438-19.
16. Schaefer JK, Rocks SS, Zheng W, Liang L, Gu B, Morel FM. 2011. Active transport, substrate specificity, and methylation of Hg(II) in anaerobic bacteria. Proc Natl Acad Sci U S A 108:8714–8719. https://doi.org/10.1073/pnas.1105781108.
17. Graham AM, Bullock AL, Maizel AC, Elias DA, Gilmour CC. 2012. Detailed assessment of the kinetics of Hg-cell association, Hg methylation, and methylmercury degradation in several *Desulfovibrio* species. Appl Environ Microbiol 78:7337–7346. https://doi.org/10.1128/AEM.01792-12.
18. Cooper CJ, Zheng K, Rush KW, Johs A, Sanders BC, Pavlopoulos GA, Kyrpides NC, Podar M, Ovchinnikov S, Ragsdale SW, Parks JM. 2020. Structure determination of the HgcAB complex using metagenome sequence data: insights into microbial mercury methylation. Commun Biol 3:320. https://doi.org/10.1038/s42003-020-1047-5.
19. Smith SD, Bridou R, Johs A, Parks JM, Elias DA, Hurt RA, Jr, Brown SD, Podar M, Wall JD. 2015. Site-directed mutagenesis of HgcA and HgcB reveals amino acid residues important for mercury methylation. Appl Environ Microbiol 81:3205–3217. https://doi.org/10.1128/AEM.00217-15.
20. Zhang Y, Rodionov DA, Gelfand MS, Gladyshev VN. 2009. Comparative genomic analyses of nickel, cobalt and vitamin B12 utilization. BMC Genomics 10:78. https://doi.org/10.1186/1471-2164-10-78.
21. Bridwell-Rabb J, Drennan CL. 2017. Vitamin B12 in the spotlight again. Curr Opin Chem Biol 37:63–70. https://doi.org/10.1016/j.cbpa.2017.01.013.
22. Goulding CW, Postigo D, Matthews RG. 1997. Cobalamin-dependent methionine synthase is a modular protein with distinct regions for binding homocysteine, methyltetrahydrofolate, cobalamin, and adenosylmethionine. Biochemistry 36:8082–8091. https://doi.org/10.1021/bi9705164.
23. Ragsdale S, Lindahl P, Münck E. 1987. Mössbauer, EPR, and optical studies of the corrinoid/iron-sulfur protein involved in the synthesis of acetyl coenzyme A by Clostridium thermoaceticum. J Biol Chem 262:14289–14297. https://doi.org/10.1016/S0021-9258(18)47936-7.
24. Choi SC, Chase T, Bartha R. 1994. Enzymatic catalysis of mercury methylation by *Desulfovibrio desulfuricans* LS. Appl Environ Microbiol 60:1342–1346. https://doi.org/10.1128/aem.60.4.1342-1346.1994.
25. Choi SC, Chase T, Bartha R. 1994. Metabolic pathways leading to mercury methylation in *Desulfovibrio desulfuricans* LS. Appl Environ Microbiol 60:4072–4077. https://doi.org/10.1128/aem.60.11.4072-4077.1994.
26. Siciliano SD, Lean DRS. 2002. Methyltransferase: an enzyme assay for microbial methylmercury formation in acidic soils and sediments. Environ Toxicol Chem 21:1184–1190. https://doi.org/10.1002/etc.5620210610.
27. Zhou J, Riccardi D, Beste A, Smith JC, Parks JM. 2014. Mercury methylation by HgcA: theory supports carbanion transfer to Hg(II). Inorg Chem 53:772–777. https://doi.org/10.1021/ic401992y.
28. Qian C, Chen H, Johs A, Lu X, An J, Pierce EM, Parks JM, Elias DA, Hettich RL, Gu B. 2018. Quantitative proteomic analysis of biological processes and responses of the bacterium Desulfovibrio desulfuricans ND132 upon deletion of its mercury methylation genes. Proteomics 18:1700479. https://doi.org/10.1002/pmic.201700479.
29. Goñi-Urriza M, Corsellis Y, Lanceleur L, Tessier E, Gury J, Monperrus M, Guyoneaud R. 2015. Relationships between bacterial energetic metabolism, mercury methylation potential, and hgcA/hgcB gene expression in Desulfovibrio dechloroacetivorans BerOc1. Environ Sci Pollut Res Int 22:13764–13771. https://doi.org/10.1007/s11356-015-4273-5.
30. Gilmour CC, Elias DA, Kucken AM, Brown SD, Palumbo AV, Schadt CW, Wall JD. 2011. Sulfate-reducing bacterium *Desulfovibrio desulfuricans* ND132 as a model for understanding bacterial mercury methylation. Appl Environ Microbiol 77:3938–3951. https://doi.org/10.1128/AEM.02993-10.
31. Grégoire DS, Poulain AJ. 2018. Shining light on recent advances in microbial mercury cycling. Facets 3:858–879. https://doi.org/10.1139/facets-2018-0015.
32. Rathore SS, Schwartz GE, Brooks SC, Painter SL. 2022. Joint estimation of biogeochemical model parameters from multiple experiments: a Bayesian approach applied to mercury methylation. Environ Model Softw 155:105453. https://doi.org/10.1016/j.envsoft.2022.105453.
33. Bravo AG, Cosio C. 2020. Biotic formation of methylmercury: a bio–physico-chemical conundrum. Limnol Oceanogr 65:1010–1027. https://doi.org/10.1002/lno.11366.
34. Hsu-Kim H, Kucharzyk KH, Zhang T, Deshusses MA. 2013. Mechanisms regulating mercury bioavailability for methylating microorganisms in the aquatic environment: a critical review. Environ Sci Technol 47:2441–2456. https://doi.org/10.1021/es304370g.
35. Christensen GA, Gionfriddo CM, King AJ, Moberly JG, Miller CL, Somenahally AC, Callister SJ, Brewer H, Podar M, Brown SD, Palumbo AV, Brandt CC, Wymore AM, Brooks SC, Hwang C, Fields MW, Wall JD, Gilmour CC, Elias DA. 2019. Determining the reliability of measuring mercury cycling gene abundance with correlations with mercury and methylmercury concentrations. Environ Sci Technol 53:8649–8663. https://doi.org/10.1021/acs.est.8b06389.
36. Schaefer JK, Kronberg RM, Björn E, Skyllberg U. 2020. Anaerobic guilds responsible for mercury methylation in boreal wetlands of varied trophic status serving as either a methylmercury source or sink. Environ Microbiol 22:3685–3699. https://doi.org/10.1111/1462-2920.15134.
37. Rocca JD, Hall EK, Lennon JT, Evans SE, Waldrop MP, Cotner JB, Nemergut DR, Graham EB, Wallenstein MD. 2015. Relationships between protein-encoding gene abundance and corresponding process are commonly assumed yet rarely observed. ISME J 9:1693–1699. https://doi.org/10.1038/ismej.2014.252.
38. Liu Y-R, Yang Z, Zhou X, Qu X, Li Z, Zhong H. 2019. Overlooked role of putative non-Hg methylators in predicting methylmercury production in paddy soils. Environ Sci Technol 53:12330–12338. https://doi.org/10.1021/acs.est.9b03013.
39. Capo E, Feng C, Bravo AG, Bertilsson S, Soerensen AL, Pinhassi J, Buck M, Karlsson C, Hawkes J, Björn E. 2022. Expression levels of hgcAB genes and mercury availability jointly explain methylmercury formation in stratified brackish waters. Environ Sci Technol 56:13119–13130. https://doi.org/10.1021/acs.est.2c03784.
40. Bouchet S, Goñi-Urriza M, Monperrus M, Guyoneaud R, Fernandez P, Heredia C, Tessier E, Gassie C, Point D, Guédron S, Achá D, Amouroux D. 2018. Linking microbial activities and low-molecular-weight thiols to Hg methylation in biofilms and periphyton from high-altitude tropical lakes in the Bolivian Altiplano. Environ Sci Technol 52:9758–9767. https://doi.org/10.1021/acs.est.8b01885.
41. Lin H, Ascher DB, Myung Y, Lamborg CH, Hallam SJ, Gionfriddo CM, Holt KE, Moreau JW. 2020. Mercury methylation by metabolically versatile and cosmopolitan marine bacteria. ISME J 15:1810–1825. https://doi.org/10.1038/s41396-020-00889-4.
42. Peterson BD, McDaniel EA, Schmidt AG, Lepak RF, Janssen SE, Tran PQ, Marick RA, Ogorek JM, DeWild JF, Krabbenhoft DP, McMahon KD. 2020. Mercury methylation genes identified across diverse anaerobic microbial guilds in a eutrophic sulfate-enriched lake. Environ Sci Technol 54:15840–15851. https://doi.org/10.1021/acs.est.0c05435.
43. Gionfriddo CM, Stott MB, Power JF, Ogorek JM, Krabbenhoft DP, Wick R, Holt K, Chen L-X, Thomas BC, Banfield JF, Moreau JW. 2020. Genome-resolved metagenomics and detailed geochemical speciation analyses yield new insights into microbial mercury cycling in geothermal springs. Appl Environ Microbiol 86:e00176-20. https://doi.org/10.1128/AEM.00176-20.
44. Leyn SA, Suvorova IA, Kholina TD, Sherstneva SS, Novichkov PS, Gelfand MS, Rodionov DA. 2014. Comparative genomics of transcriptional regulation of methionine metabolism in proteobacteria. PLoS One 9:e113714. https://doi.org/10.1371/journal.pone.0113714.
45. Novichkov PS, Li X, Kuehl JV, Deutschbauer AM, Arkin AP, Price MN, Rodionov DA. 2014. Control of methionine metabolism by the SahR transcriptional regulator in Proteobacteria. Environ Microbiol 16:1–8. https://doi.org/10.1111/1462-2920.12273.
46. Li X, Krumholz LR. 2007. Regulation of arsenate resistance in *Desulfovibrio desulfuricans* G20 by an *arsRBCC* operon and an *arsC* gene. J Bacteriol 189:3705–3711. https://doi.org/10.1128/JB.01913-06.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.

47. Ekstrom EB, Morel FMM. 2008. Cobalt limitation of growth and mercury methylation in sulfate-reducing bacteria. Environ Sci Technol 42:93–99. https://doi.org/10.1021/es0705644.
48. Smith SD. 2015. Protein components of the microbial mercury methylation pathway. PhD dissertation. University of Missouri, Columbia, MO.
49. Busenlehner LS, Pennella MA, Giedroc DP. 2003. The SmtB/ArsR family of metalloregulatory transcriptional repressors: structural insights into prokaryotic metal resistance. FEMS Microbiol Rev 27:131–143. https://doi.org/10.1016/S0168-6445(03)00054-8.
50. Andres J, Bertin PN. 2016. The microbial genomics of arsenic. FEMS Microbiol Rev 40:299–322. https://doi.org/10.1093/femsre/fuv050.
51. Chen J, Madegowda M, Bhattacharjee H, Rosen BP. 2015. ArsP: a methylarsenite efflux permease. Mol Microbiol 98:625–635. https://doi.org/10.1111/mmi.13145.
52. Chen J, Zhang J, Rosen BP. 2019. Role of ArsEFG in roxarsone and nitarsone detoxification and resistance. Environ Sci Technol 53:6182–6191. https://doi.org/10.1021/acs.est.9b01187.
53. Rawle R, Saley TC, Kang Y-S, Wang Q, Walk S, Bothner B, McDermott TR. 2021. Introducing the ArsR-regulated arsenic stimulon. Front Microbiol 12:630562. https://doi.org/10.3389/fmicb.2021.630562.
54. Qin J, Rosen BP, Zhang Y, Wang G, Franke S, Rensing C. 2006. Arsenic detoxification and evolution of trimethylarsine gas by a microbial arsenite S-adenosylmethionine methyltransferase. Proc Natl Acad Sci U S A 103:2075–2080. https://doi.org/10.1073/pnas.0506836103.
55. Woodcroft BJ, Singleton CM, Boyd JA, Evans PN, Emerson JB, Zayed AAF, Hoelzle RD, Lamberton TO, McCalley CK, Hodgkins SB, Wilson RM, Purvine SO, Nicora CD, Li C, Frolking S, Chanton JP, Crill PM, Saleska SR, Rich VI, Tyson GW. 2018. Genome-centric view of carbon processing in thawing permafrost. Nature 560:49–54. https://doi.org/10.1038/s41586-018-0338-1.
56. Haas S, Desai DK, LaRoche J, Pawlowicz R, Wallace DWR. 2019. Geomicrobiology of the carbon, nitrogen and sulphur cycles in Powell Lake: a permanently stratified water column containing ancient seawater. Environ Microbiol 21:3927–3952. https://doi.org/10.1111/1462-2920.14743.
57. Qiu YL, Kuang XZ, Shi XS, Yuan XZ, Guo RB. 2014. Paludibacter jiangxiensis sp. nov., a strictly anaerobic, propionate-producing bacterium isolated from rice paddy field. Arch Microbiol 196:149–155. https://doi.org/10.1007/s00203-013-0951-1.
58. Anantharaman K, Brown CT, Hug LA, Sharon I, Castelle CJ, Probst AJ, Thomas BC, Singh A, Wilkins MJ, Karaoz U, Brodie EL, Williams KH, Hubbard SS, Banfield JF. 2016. Thousands of microbial genomes shed light on interconnected biogeochemical processes in an aquifer system. Nat Commun 7:13219. https://doi.org/10.1038/ncomms13219.
59. Giloteaux L, Holmes DE, Williams KH, Wrighton KC, Wilkins MJ, Montgomery AP, Smith JA, Orellana R, Thompson CA, Roper TJ, Long PE, Lovley DR. 2013. Characterization and transcription of arsenic respiration and resistance genes during in situ uranium bioremediation. ISME J 7:370–383. https://doi.org/10.1038/ismej.2012.109.
60. Prabaharan C, Kandavelu P, Packianathan C, Rosen BP, Thiyagarajan S. 2019. Structures of two ArsR As(III)-responsive transcriptional repressors: implications for the mechanism of derepression. J Struct Biol 207:209–217. https://doi.org/10.1016/j.jsb.2019.05.009.
61. Mirdita M, Schütze K, Moriwaki Y, Heo L, Ovchinnikov S, Steinegger M. 2022. ColabFold: making protein folding accessible to all. Nat Methods 19:679–682. https://doi.org/10.1038/s41592-022-01488-1.
62. Jumper J, Evans R, Pritzel A, Green T, Figurnov M, Ronneberger O, Tunyasuvunakool K, Bates R, Žídek A, Potapenko A, Bridgland A, Meyer C, Kohl SAA, Ballard AJ, Cowie A, Romera-Paredes B, Nikolov S, Jain R, Adler J, Back T, Petersen S, Reiman D, Clancy E, Zielinski M, Steinegger M, Pacholska M, Berghammer T, Bodenstein S, Silver D, Vinyals O, Senior AW, Kavukcuoglu K, Kohli P, Hassabis D. 2021. Highly accurate protein structure prediction with AlphaFold. Nature 596:583–589. https://doi.org/10.1038/s41586-021-03819-2.
63. Shi W, Dong J, Scott RA, Ksenzenko MY, Rosen BP. 1996. The role of arsenic-thiol interactions in metalloregulation of the ars operon. J Biol Chem 271:9291–9297. https://doi.org/10.1074/jbc.271.16.9291.
64. Packianathan C, Kandavelu P, Rosen BP. 2018. The structure of an As(III) S-adenosylmethionine methyltransferase with 3-coordinately bound As(III) depicts the first step in catalysis. Biochemistry 57:4083–4092. https://doi.org/10.1021/acs.biochem.8b00457.
65. De Francisco P, Martín-González A, Rodriguez-Martín D, Díaz S. 2021. Interactions with arsenic: mechanisms of toxicity and cellular resistance in eukaryotic microorganisms. Int J Environ Res Public Health 18:12226. https://doi.org/10.3390/ijerph182212226.
66. Rush KW. 2018. Expression and characterization of HgcA and HgcB, two proteins involved in methylmercury biosynthesis. PhD dissertation. University of Michigan, Ann Arbor, MI.
67. Barrouilhet S, Monperrus M, Tessier E, Khalfaoui-Hassani B, Guyoneaud R, Isaure M-P, Goñi-Urriza M. 2023. Effect of exogenous and endogenous sulfide on the production and the export of methylmercury by sulfate reducing bacteria. Environ Sci Pollut Res Int 30:3835–3846. https://doi.org/10.1007/s11356-022-22173-y.
68. Benoit JM, Gilmour CC, Mason RP. 2001. The influence of sulfide on solid phase mercury bioavailability for methylation by pure cultures of Desulfobulbus propionicus (1pr3). Environ Sci Technol 35:127–132. https://doi.org/10.1021/es001415n.
69. Graham AM, Aiken GR, Gilmour CC. 2012. Dissolved organic matter enhances microbial mercury methylation under sulfidic conditions. Environ Sci Technol 46:2715–2723. https://doi.org/10.1021/es203658f.
70. Aždajić M, Blais JM, Poulain AJ. 2022. Arsenate decreases production of methylmercury across increasing sulfate concentration amendments in freshwater lake sediments. Environ Sci Process Impacts 24:1508–1516. https://doi.org/10.1039/D1EM00543J.
71. Glasser NR, Oyala PH, Osborne TH, Santini JM, Newman DK. 2018. Structural and mechanistic analysis of the arsenate respiratory reductase provides insight into environmental arsenic transformations. Proc Natl Acad Sci U S A 115:E8614–E8623. https://doi.org/10.1073/pnas.1807984115.
72. Stolz JF, Basu P. 2018. Unraveling the inner workings of respiratory arsenate reductase. Proc Natl Acad Sci U S A 115:9051–9053. https://doi.org/10.1073/pnas.1812841115.
73. Malasarn D, Keeffe JR, Newman DK. 2008. Characterization of the arsenate respiratory reductase from *Shewanella* sp. strain ANA-3. J Bacteriol 190:135–142. https://doi.org/10.1128/JB.01110-07.
74. Tsuchiya T, Ehara A, Kasahara Y, Hamamura N, Amachi S, Zhou N-Y. 2019. Expression of genes and proteins involved in arsenic respiration and resistance in dissimilatory arsenate-reducing *Geobacter* sp. strain OR-1. Appl Environ Microbiol 85:e00763-19. https://doi.org/10.1128/AEM.00763-19.
75. Richey C, Chovanec P, Hoeft SE, Oremland RS, Basu P, Stolz JF. 2009. Respiratory arsenate reductase as a bidirectional enzyme. Biochem Biophys Res Commun 382:298–302. https://doi.org/10.1016/j.bbrc.2009.03.045.
76. Jormakka M, Yokoyama K, Yano T, Tamakoshi M, Akimoto S, Shimamura T, Curmi P, Iwata S. 2008. Molecular mechanism of energy conservation in polysulfide respiration. Nat Struct Mol Biol 15:730–737. https://doi.org/10.1038/nsmb.1434.
77. Wang S, Jiang L, Hu Q, Liu X, Yang S, Shao Z. 2021. Elemental sulfur reduction by a deep-sea hydrothermal vent Campylobacterium Sulfurimonas sp. NW10. Environ Microbiol 23:965–979. https://doi.org/10.1111/1462-2920.15247.
78. Jay ZJ, Beam JP, Dohnalkova A, Lohmayer R, Bodle B, Planer-Friedrich B, Romine M, Inskeep WP. 2015. *Pyrobaculum yellowstonensis* strain WP30 respires on elemental sulfur and/or arsenate in circumneutral sulfidic geothermal sediments of Yellowstone National Park. Appl Environ Microbiol 81:5907–5916. https://doi.org/10.1128/AEM.01095-15.
79. Oppermann FB, Schmidt B, Steinbüchel A. 1991. Purification and characterization of acetoin:2,6-dichlorophenolindophenol oxidoreductase, dihydrolipoamide dehydrogenase, and dihydrolipoamide acetyltransferase of the *Pelobacter carbinolicus* acetoin dehydrogenase enzyme system. J Bacteriol 173:757–767. https://doi.org/10.1128/jb.173.2.757-767.1991.
80. Oppermann FB, Steinbüchel A. 1994. Identification and molecular characterization of the *aco* genes encoding the *Pelobacter carbinolicus* acetoin dehydrogenase enzyme system. J Bacteriol 176:469–485. https://doi.org/10.1128/jb.176.2.469-485.1994.
81. de Kok A, Hengeveld AF, Martin A, Westphal AH. 1998. The pyruvate dehydrogenase multi-enzyme complex from Gram-negative bacteria. Biochim Biophys Acta 1385:353–366. https://doi.org/10.1016/s0167-4838(98)00079-x.
82. Brown SD, Gilmour CC, Kucken AM, Wall JD, Elias DA, Brandt CC, Podar M, Chertkov O, Held B, Bruce DC, Detter JC, Tapia R, Han CS, Goodwin LA, Cheng JF, Pitluck S, Woyke T, Mikhailova N, Ivanova NN, Han J, Lucas S, Lapidus AL, Land ML, Hauser LJ, Palumbo AV. 2011. Genome sequence of the mercury-methylating strain Desulfovibrio desulfuricans ND132. J Bacteriol 193:2078–2079. https://doi.org/10.1128/JB.00170-11.
83. Walker CB, He Z, Yang ZK, Ringbauer JA, He Q, Zhou J, Voordouw G, Wall JD, Arkin AP, Hazen TC, Stolyar S, Stahl DA. 2009. The electron transfer system of syntrophically grown Desulfovibrio vulgaris. J Bacteriol 191:5793–5801. https://doi.org/10.1128/JB.00356-09.
84. Arkin AP, Cottingham RW, Henry CS, Harris NL, Stevens RL, Maslov S, Dehal P, Ware D, Perez F, Canon S, Sneddon MW, Henderson ML, Riehl WJ, Murphy-Olson D, Chan SY, Kamimura RT, Kumari S, Drake MM, Brettin

TS, Glass EM, Chivian D, Gunter D, Weston DJ, Allen BH, Baumohl J, Best AA, Bowen B, Brenner SE, Bun CC, Chandonia J-M, Chia J-M, Colasanti R, Conrad N, Davis JJ, Davison BH, DeJongh M, Devoid S, Dietrich E, Dubchak I, Edirisinghe JN, Fang G, Faria JP, Frybarger PM, Gerlach W, Gerstein M, Greiner A, Gurtowski J, Haun HL, He F, Jain R, Joachimiak MP, Keegan KP, Kondo S, Kumar V, Land ML, Meyer F, Mills M, Novichkov PS, Oh T, Olsen GJ, Olson R, Parrello B, Pasternak S, Pearson E, Poon SS, Price GA, Ramakrishnan S, Ranjan P, Ronald PC, Schatz MC, Seaver SMD, Shukla M, Sutormin RA, Syed MH, Thomason J, Tintle NL, Wang D, Xia F, Yoo H, Yoo S, Yu D. 2018. KBase: The United States Department of Energy Systems Biology Knowledgebase. Nat Biotechnol 36:566–569. https://doi.org/10.1038/nbt.4163.

85. Bolger AM, Lohse M, Usadel B. 2014. Trimmomatic: a flexible trimmer for Illumina sequence data. Bioinformatics 30:2114–2120. https://doi.org/10.1093/bioinformatics/btu170.
86. Kim D, Paggi JM, Park C, Bennett C, Salzberg SL. 2019. Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype. Nat Biotechnol 37:907–915. https://doi.org/10.1038/s41587-019-0201-4.
87. Pertea M, Pertea GM, Antonescu CM, Chang T-C, Mendell JT, Salzberg SL. 2015. StringTie enables improved reconstruction of a transcriptome from RNA-seq reads. Nat Biotechnol 33:290–295. https://doi.org/10.1038/nbt.3122.
88. Love MI, Huber W, Anders S. 2014. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol 15:550. https://doi.org/10.1186/s13059-014-0550-8.
89. Afgan E, Baker D, Batut B, van den Beek M, Bouvier D, Čech M, Chilton J, Clements D, Coraor N, Grüning BA, Guerler A, Hillman-Jackson J, Hiltemann S, Jalili V, Rasche H, Soranzo N, Goecks J, Taylor J, Nekrutenko A, Blankenberg D. 2018. The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2018 update. Nucleic Acids Res 46:W537–W544. https://doi.org/10.1093/nar/gky379.
90. Gionfriddo C, Soren A, Wymore AM, Hartnett DS, Podar M, Parks JM, Elias DA, Gilmour C. 2023. RNA sequencing dataset from Pseudodesulfovibrio mercurii ND132 washed-cell cultures. Smithsonian Environmental Research Center, Edgewater, MD. https://doi.org/10.25573/serc.21561387.v1.
91. Gionfriddo C, Soren A, Wymore AM, Hartnett DS, Podar M, Parks JM, Elias DA, Gilmour C. 2023. RNA sequencing dataset from Pseudodesulfovibrio mercurii ND132 batch cultures. Smithsonian Environmental Research Center, Edgewater, MD. https://doi.org/10.25573/serc.21561189.v1.
92. Mirdita M, Steinegger M, Söding J. 2019. MMseqs2 desktop and local web server app for fast, interactive sequence searches. Bioinformatics 35:2856–2858. https://doi.org/10.1093/bioinformatics/bty1057.
93. Mitchell AL, Almeida A, Beracochea M, Boland M, Burgin J, Cochrane G, Crusoe MR, Kale V, Potter SC, Richardson LJ, Sakharova E, Scheremetjew M, Korobeynikov A, Shlemov A, Kunyavskaya O, Lapidus A, Finn RD. 2020. MGnify: the microbiome analysis resource in 2020. Nucleic Acids Res 48:D570–D578. https://doi.org/10.1093/nar/gkz1035.
94. Evans R, O'Neill M, Pritzel A, Antropova N, Senior A, Green T, Žídek A, Bates R, Blackwell S, Yim J, Ronneberger O, Bodenstein S, Zielinski M, Bridgland A, Potapenko A, Cowie A, Tunyasuvunakool K, Jain R, Clancy E, Kohli P, Jumper J, Hassabis D. 2022. Protein complex prediction with AlphaFold-Multimer. bioRxiv. https://doi.org/10.1101/2021.10.04.463034.
95. R Core Team. 2013. R: a language and environment for statistical computing.
96. Harrison KJ, Crécy-Lagard V, Zallot R. 2018. Gene Graphics: a genomic neighborhood data visualization web application. Bioinformatics 34:1406–1408. https://doi.org/10.1093/bioinformatics/btx793.
97. Sprouffske K, Wagner A. 2016. Growthcurver: an R package for obtaining interpretable metrics from microbial growth curves. BMC Bioinformatics 17:172. https://doi.org/10.1186/s12859-016-1016-7.

Downloaded from https://journals.asm.org/journal/aem on 18 October 2026 by 36.230.0.41.